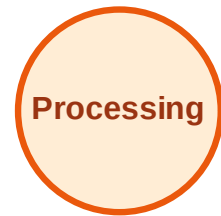


BFS Traversal

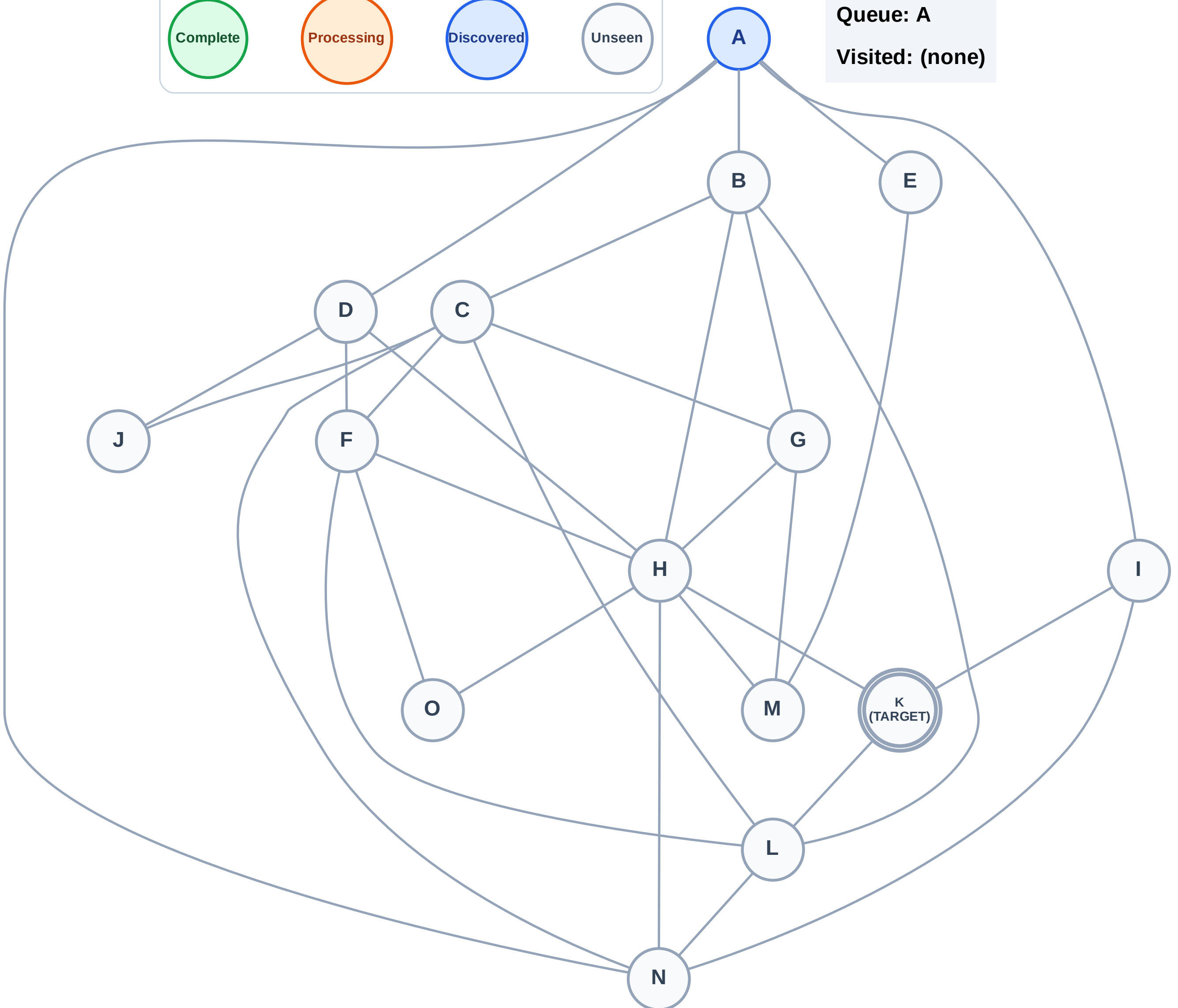
Step 1: Start at A (Seeking target node K)

Legend



Queue: A

Visited: (none)



BFS Traversal

Step 2: Process node A

Legend

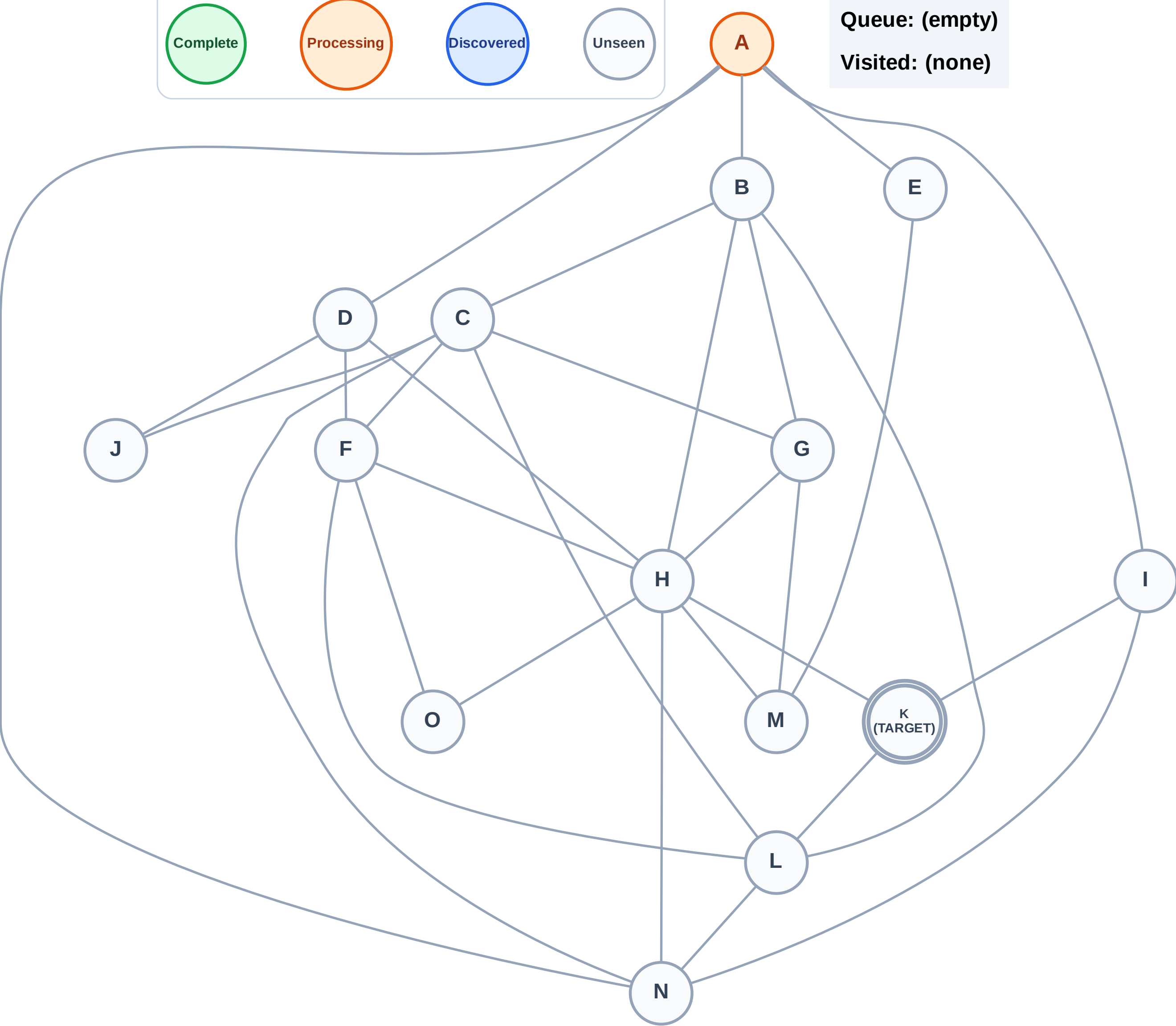
Complete

Processing

Discovered

Unseen

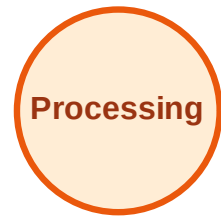
Queue: (empty)
Visited: (none)



BFS Traversal

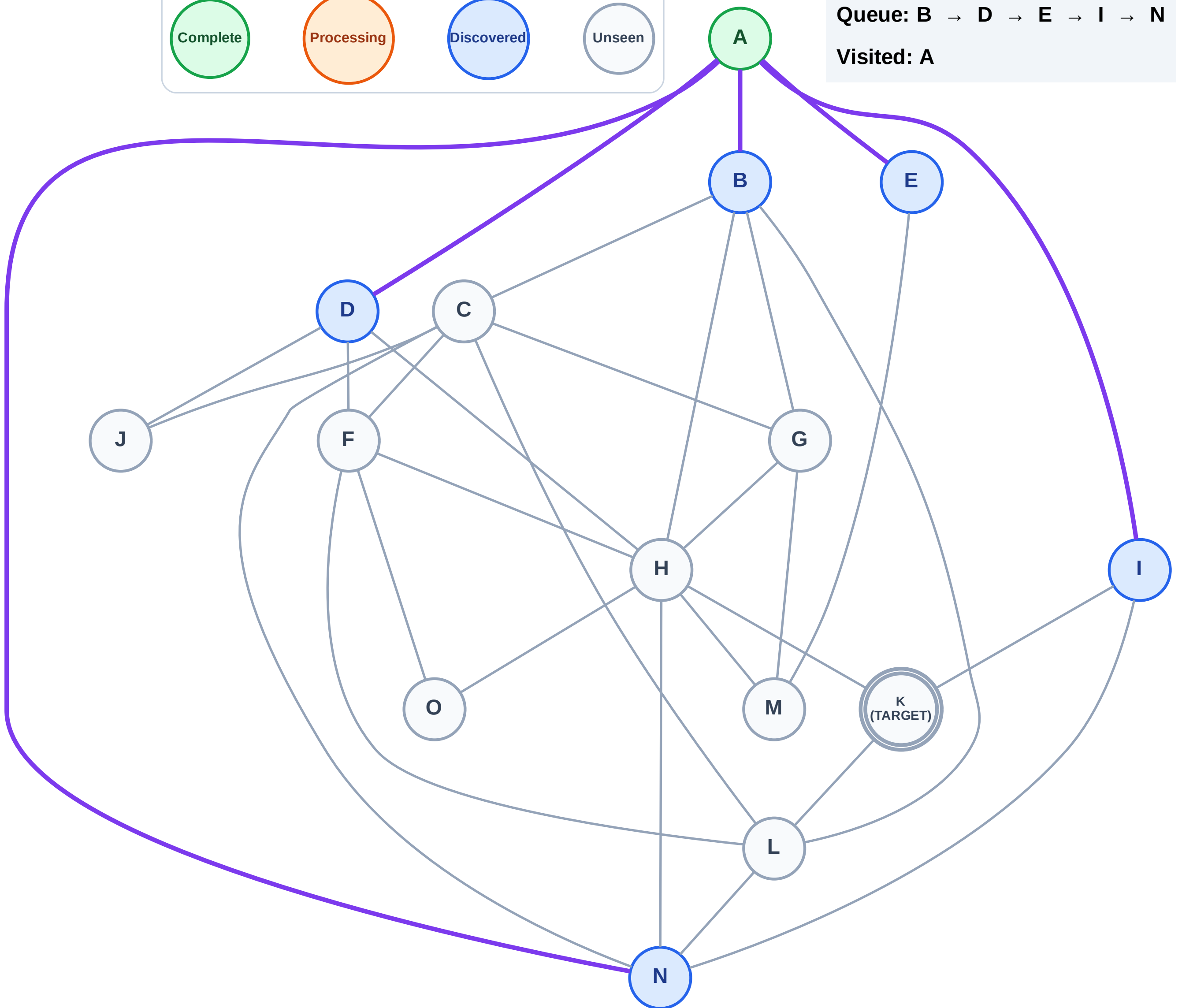
Step 3: Discover B, D, E, I, N from A

Legend



Queue: B → D → E → I → N

Visited: A



BFS Traversal

Step 4: Process node B

Legend

Complete

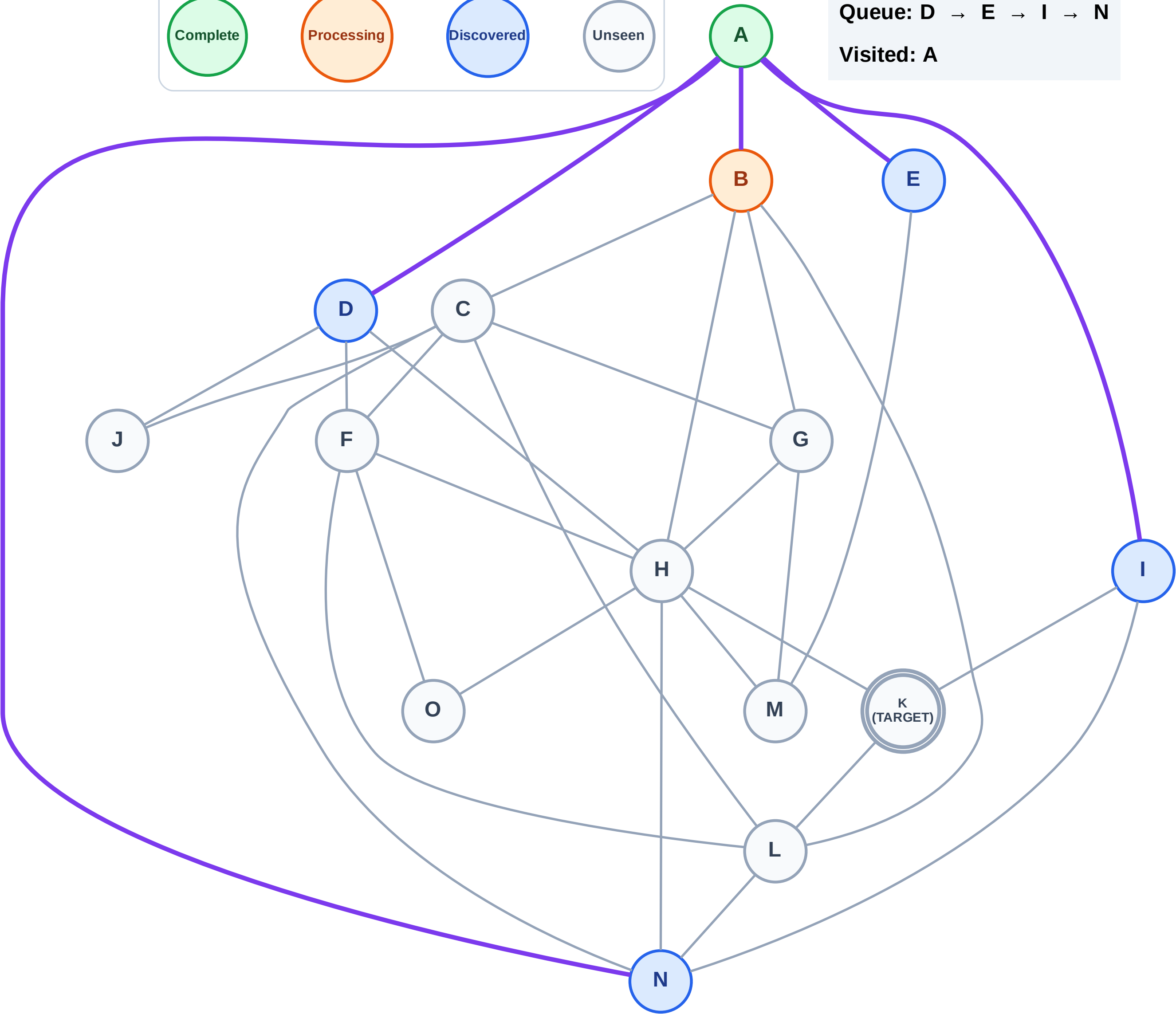
Processing

Discovered

Unseen

Queue: D → E → I → N

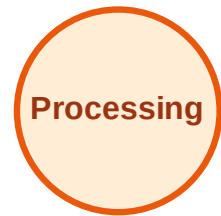
Visited: A



BFS Traversal

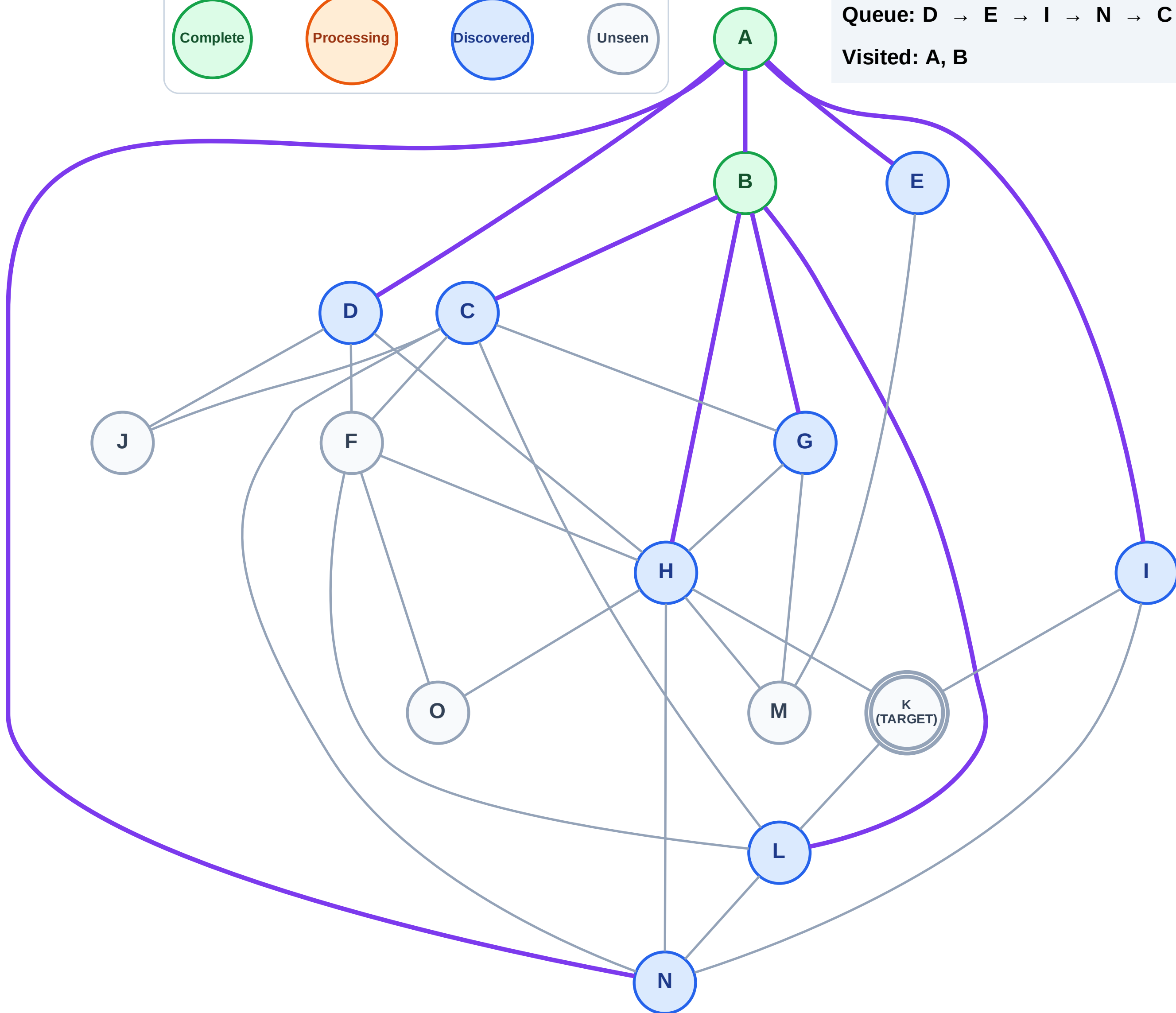
Step 5: Discover C, G, H, L from B

Legend



Queue: D → E → I → N → C → G → H → L

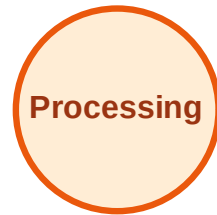
Visited: A, B



BFS Traversal

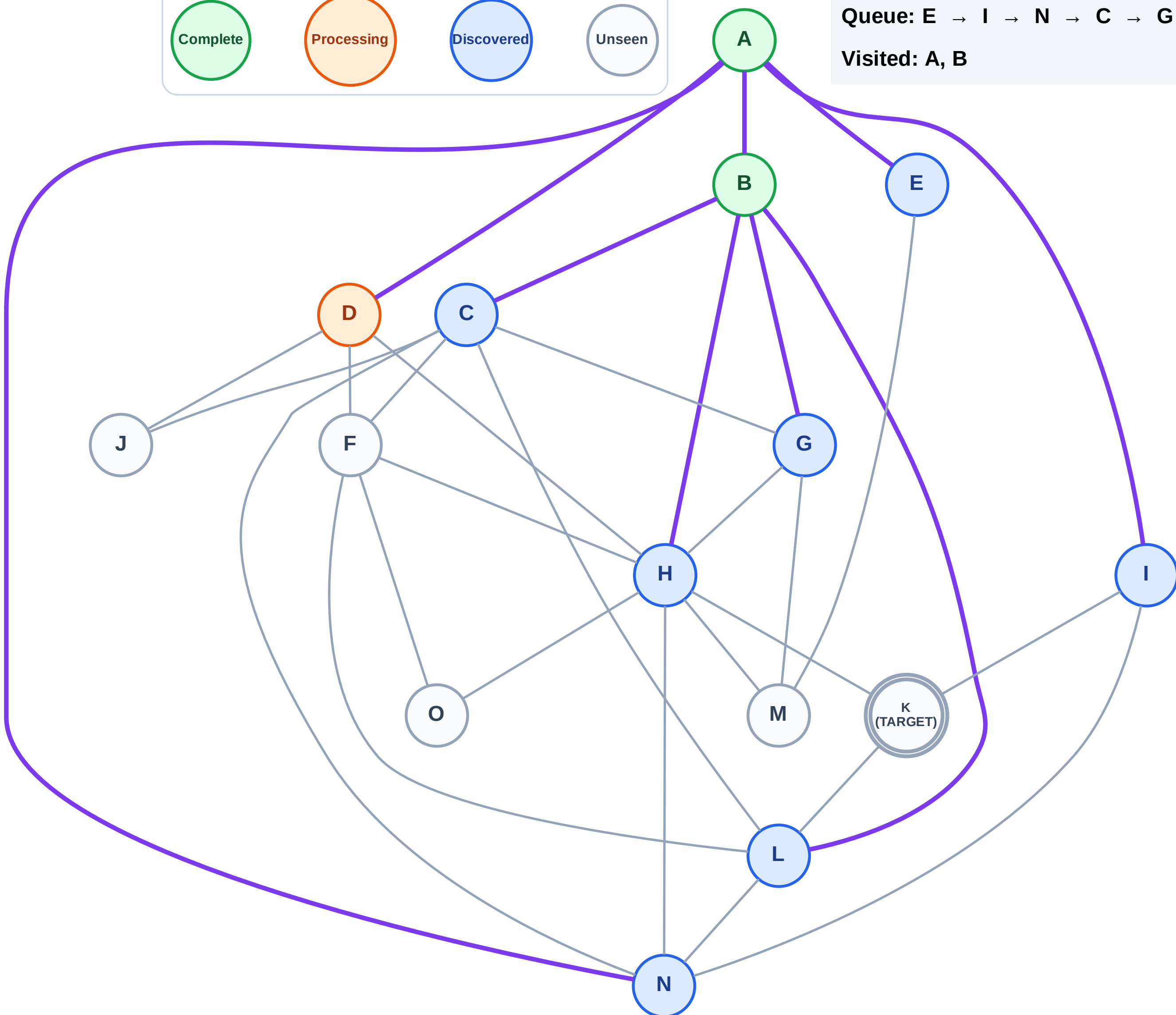
Step 6: Process node D

Legend



Queue: E → I → N → C → G → H → L

Visited: A, B



BFS Traversal

Step 7: Discover F, J from D

Legend

Complete

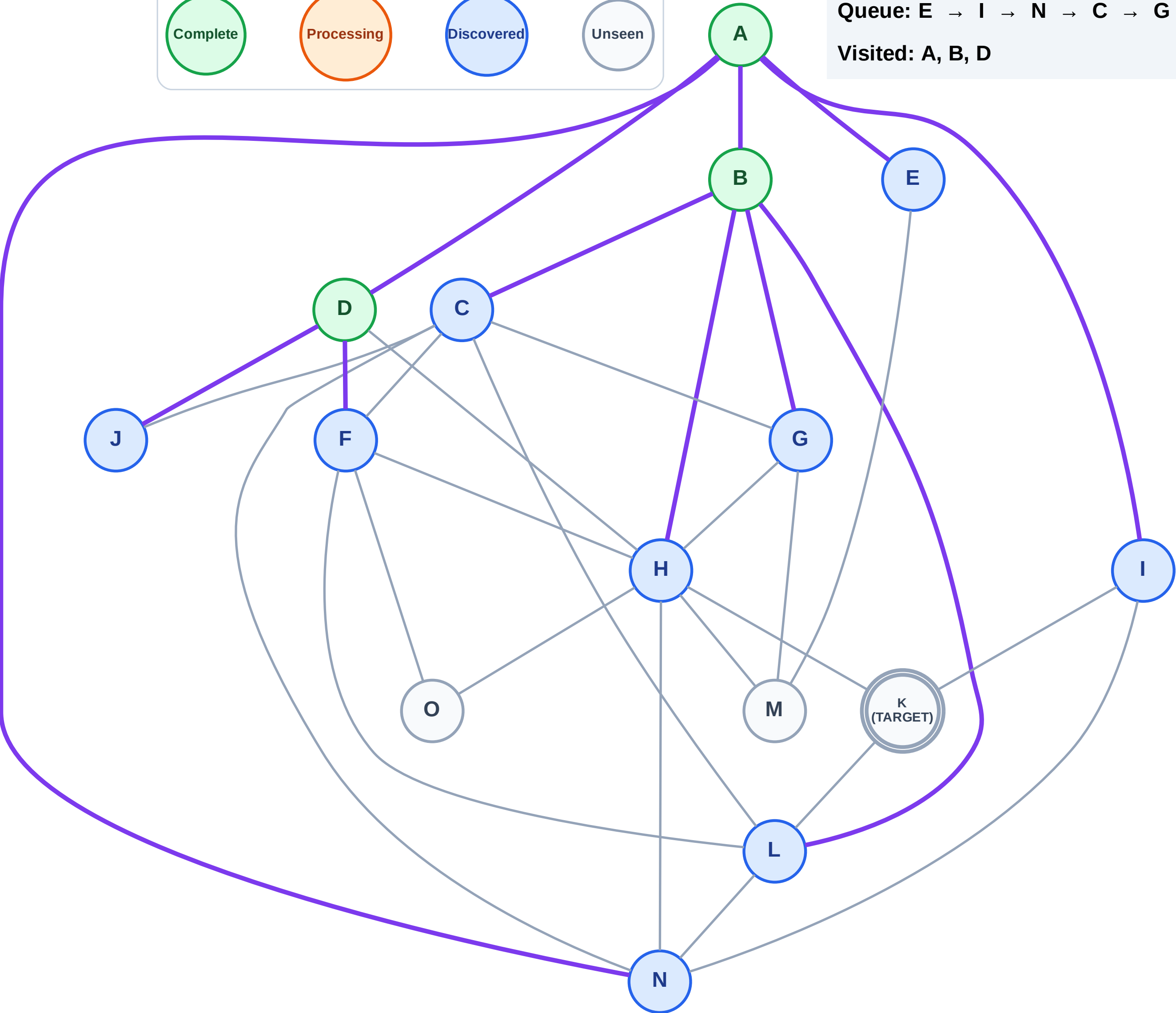
Processing

Discovered

Unseen

Queue: E → I → N → C → G → H → L → F → J

Visited: A, B, D



BFS Traversal

Step 8: Process node E

Legend

Complete

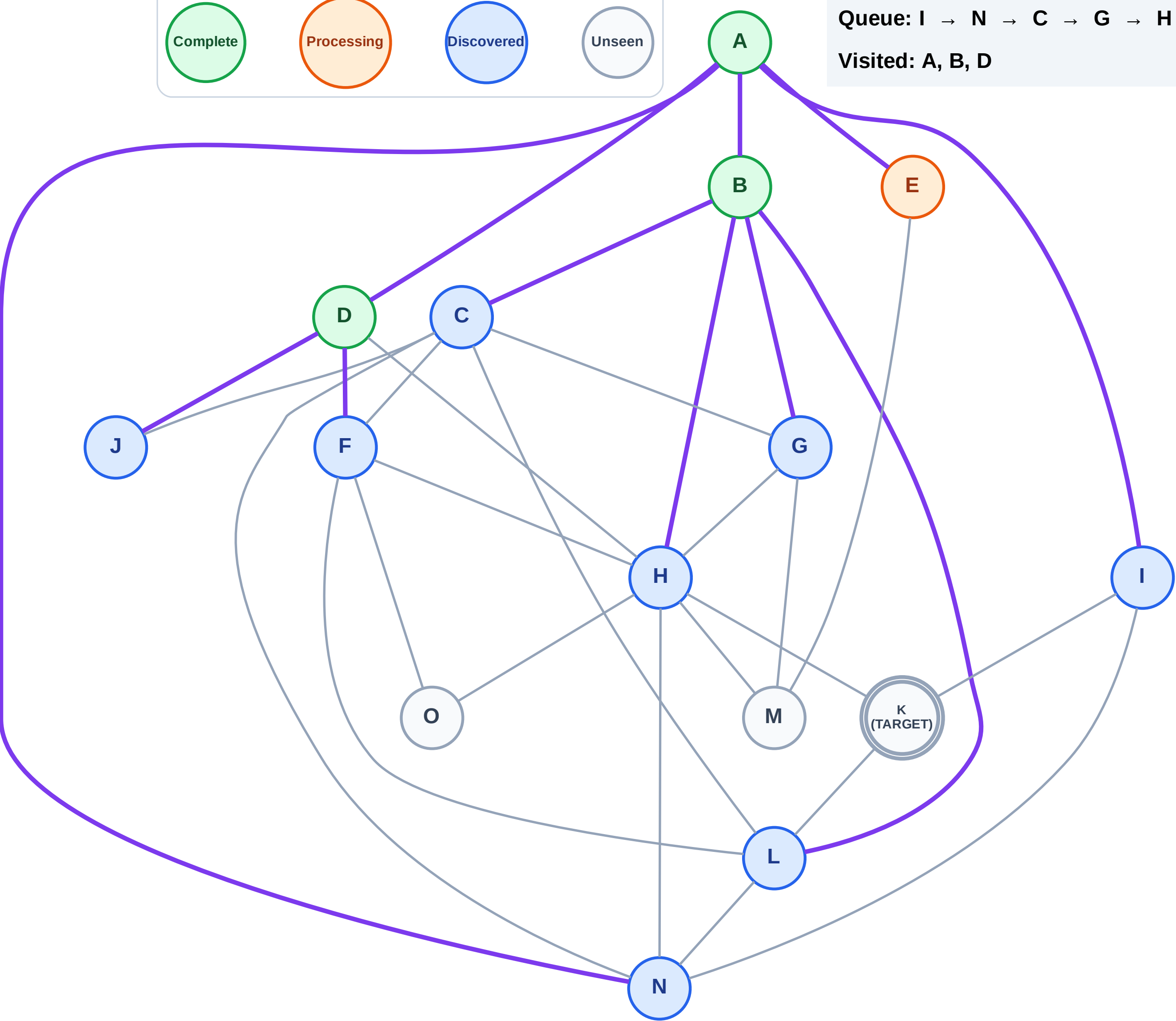
Processing

Discovered

Unseen

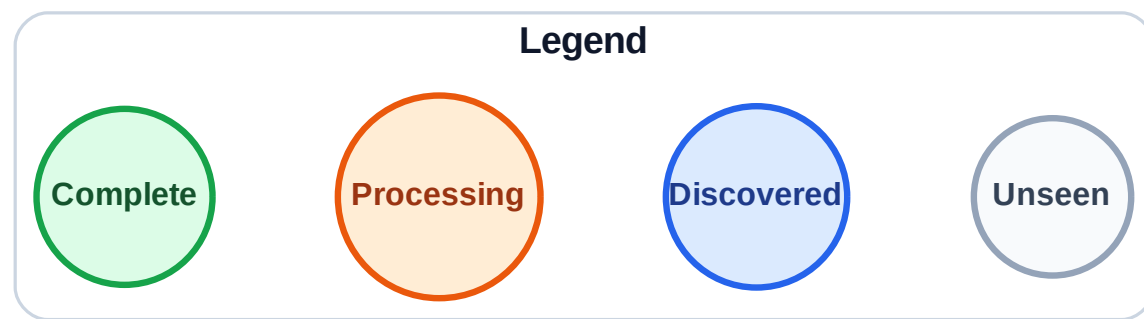
Queue: I → N → C → G → H → L → F → J

Visited: A, B, D

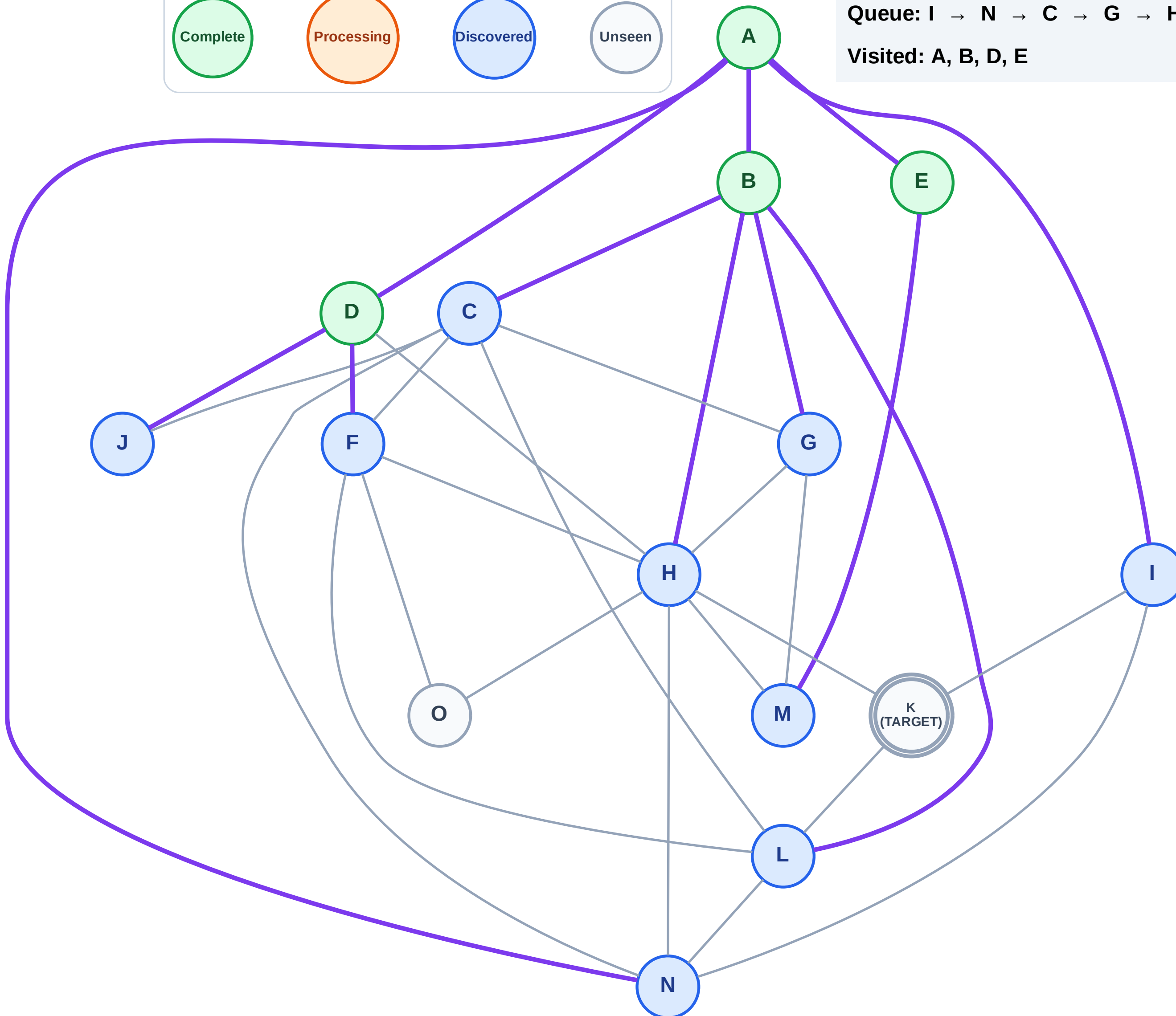


BFS Traversal

Step 9: Discover M from E



Queue: I → N → C → G → H → L → F → J → M
Visited: A, B, D, E



Step 10: Process node I

Step 10: Process node I

Complete



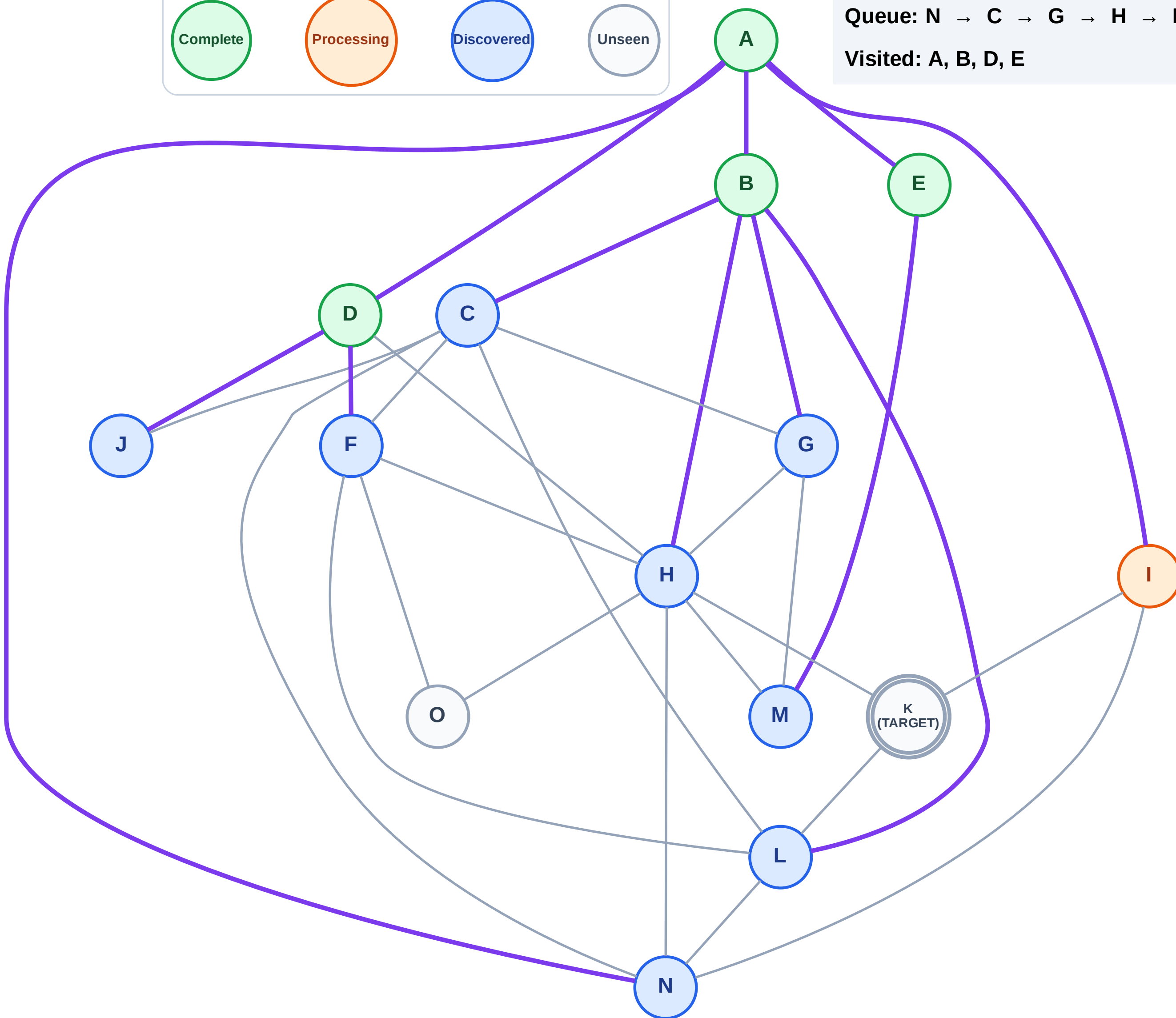
Processing



Discovered

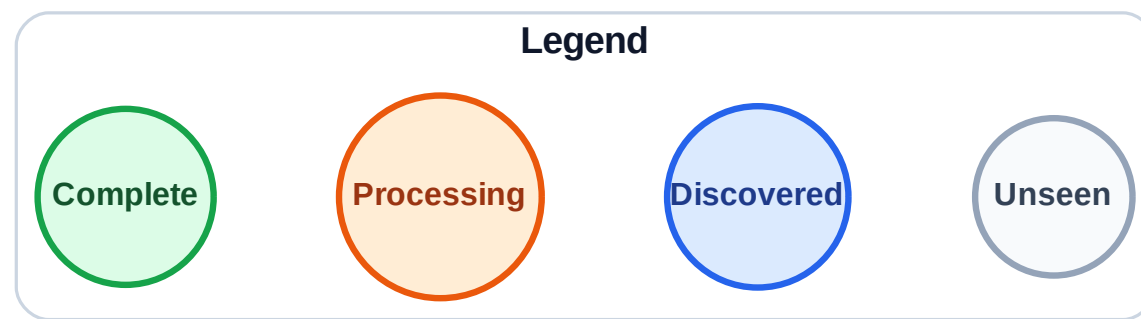
Unseen

Visited: A, B, D, E

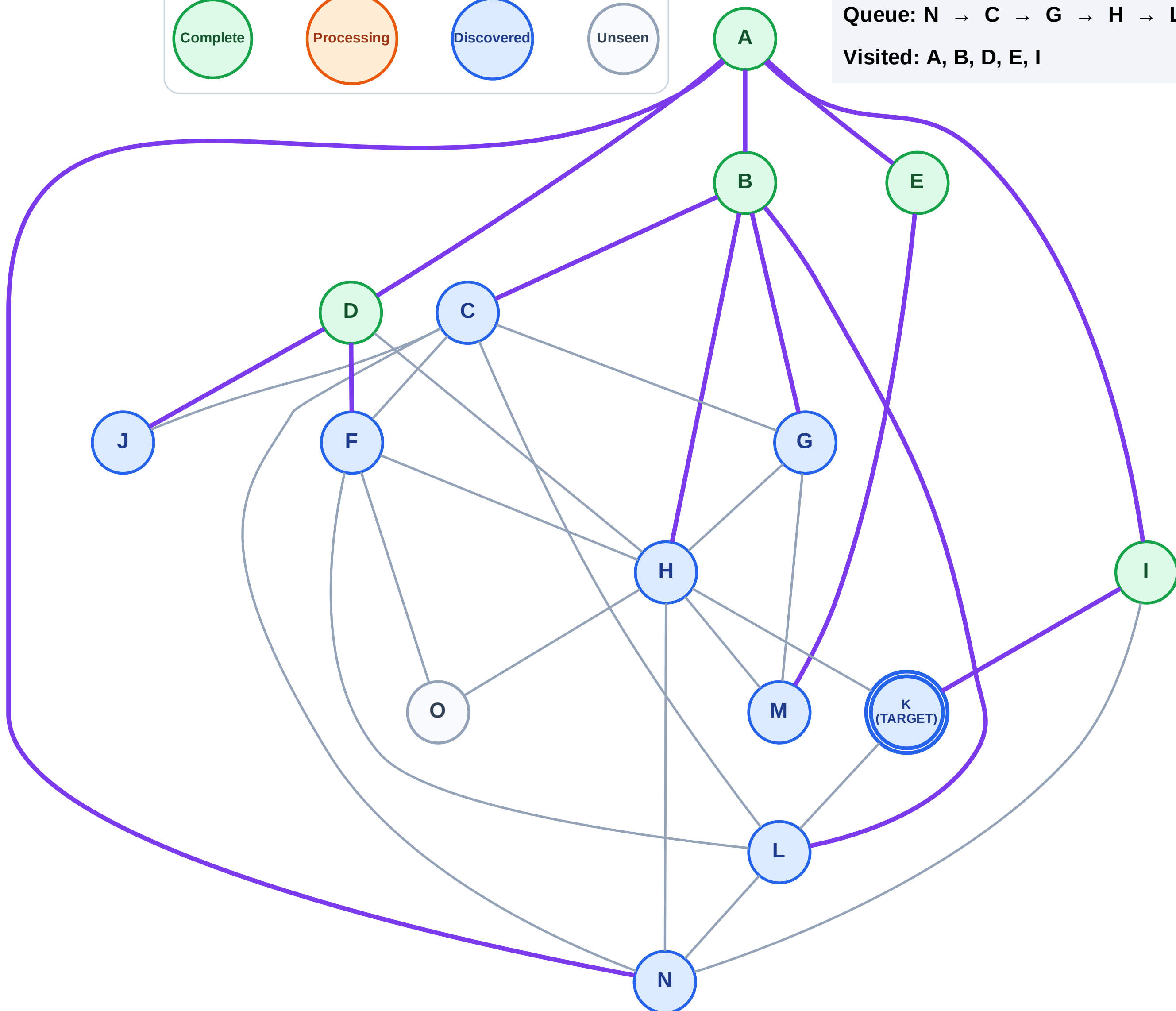


BFS Traversal

Step 11: Discover K from I

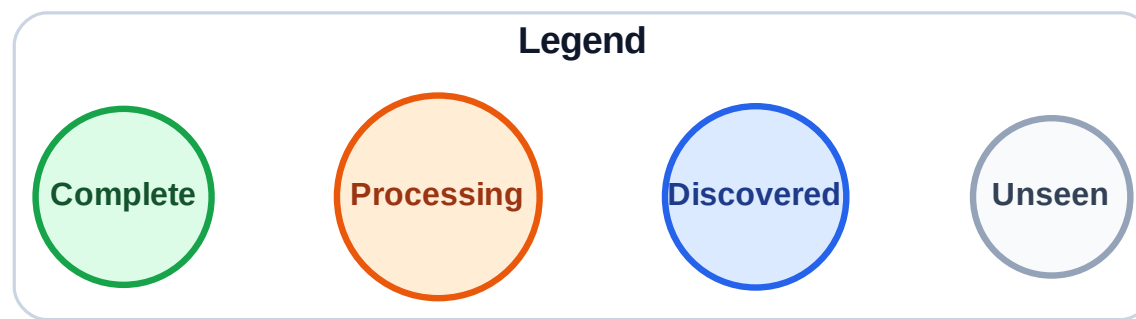


Queue: N → C → G → H → L → F → J → M → K
Visited: A, B, D, E, I

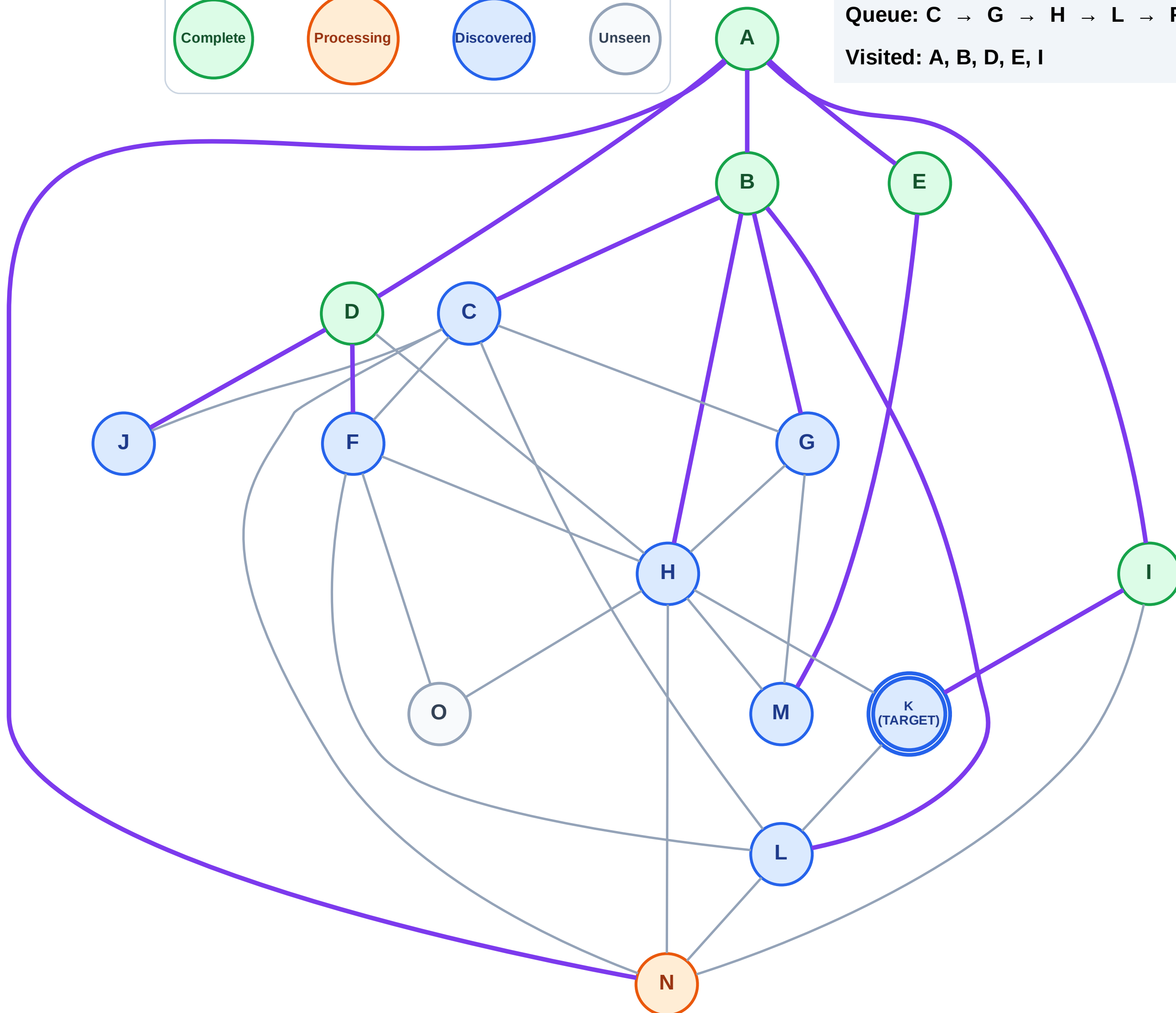


BFS Traversal

Step 12: Process node N



Queue: C → G → H → L → F → J → M → K
Visited: A, B, D, E, I



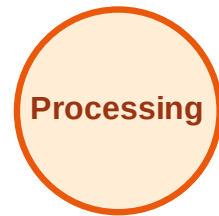
BFS Traversal

Step 13: No new neighbors from N

Legend



Complete



Processing



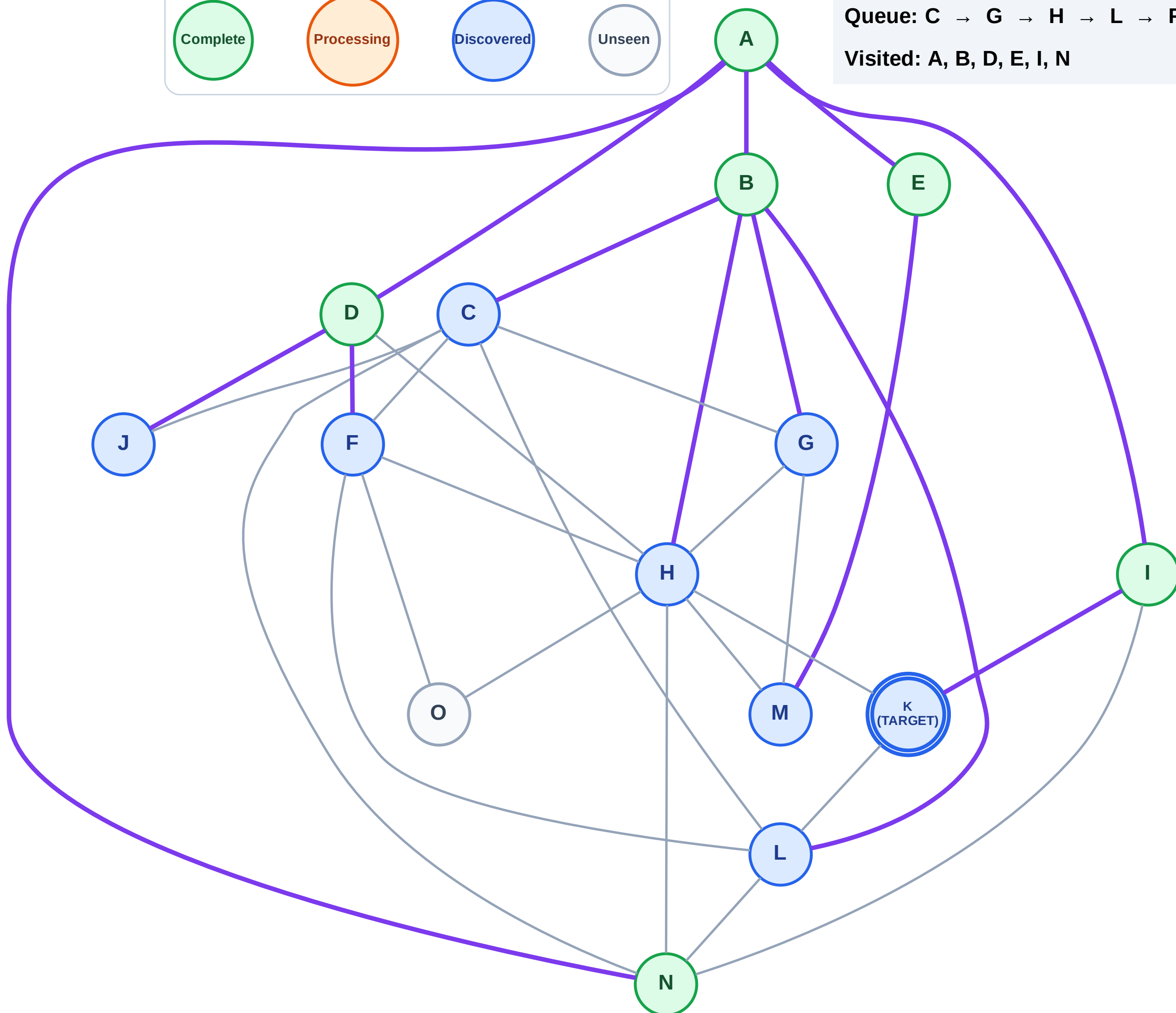
Discovered



Unseen

Queue: C → G → H → L → F → J → M → K

Visited: A, B, D, E, I, N



BFS Traversal

Step 14: Process node C

Legend

Complete

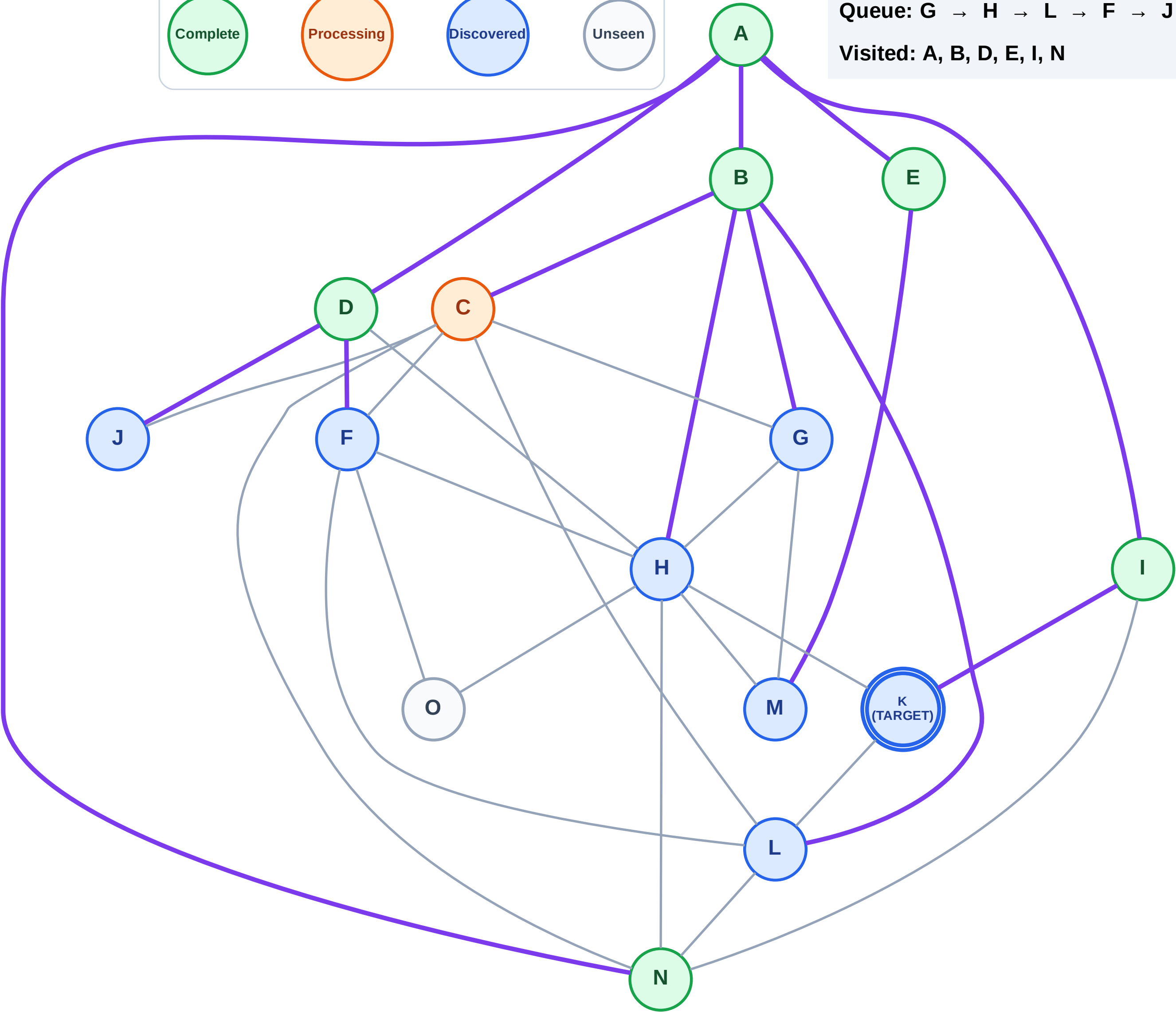
Processing

Discovered

Unseen

Queue: G → H → L → F → J → M → K

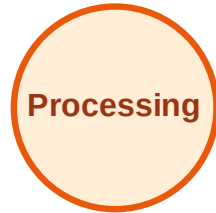
Visited: A, B, D, E, I, N



BFS Traversal

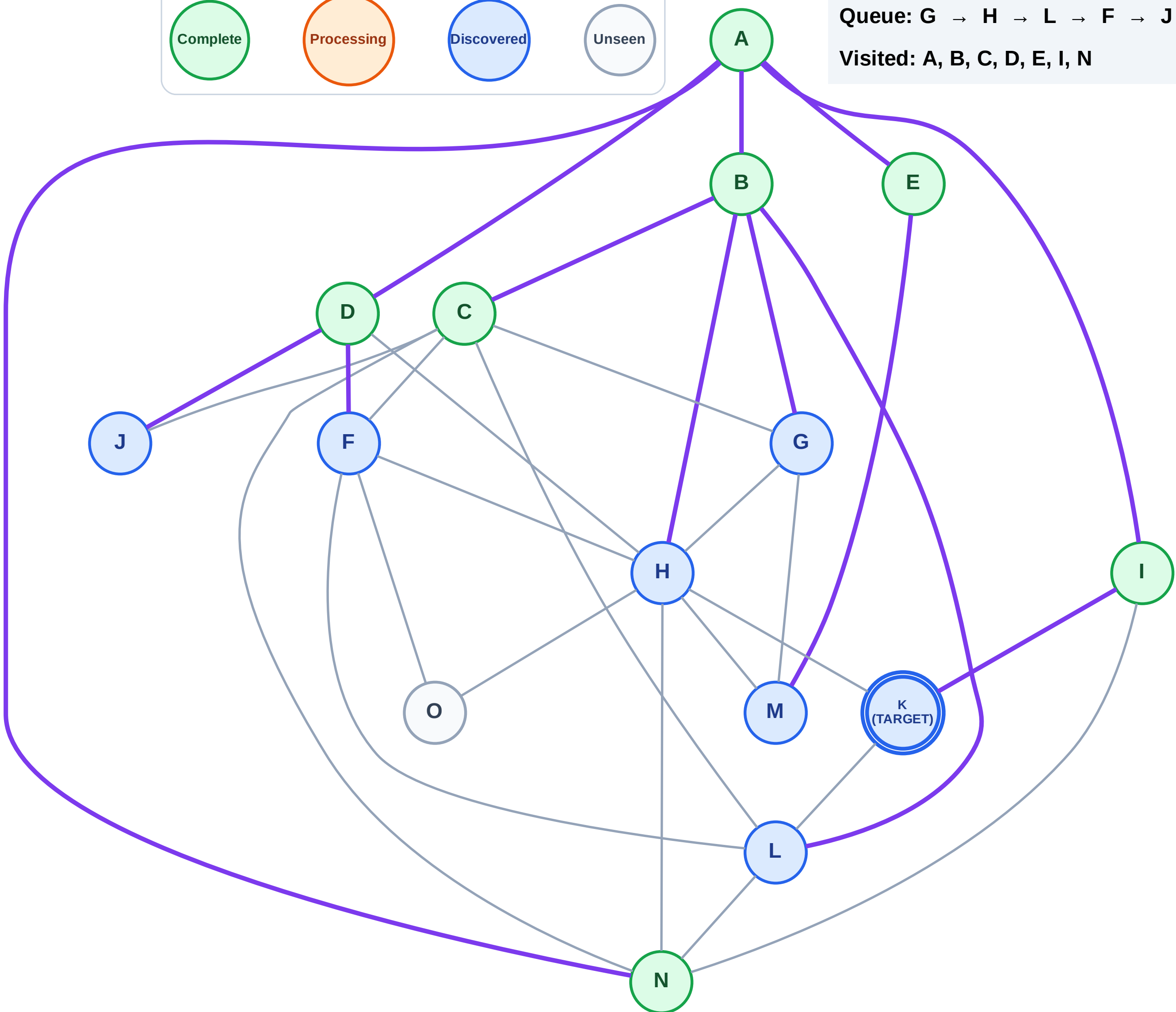
Step 15: No new neighbors from C

Legend



Queue: G → H → L → F → J → M → K

Visited: A, B, C, D, E, I, N



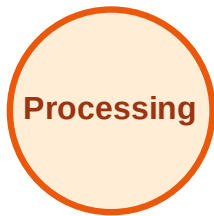
BFS Traversal

Step 16: Process node G

Legend



Complete



Processing



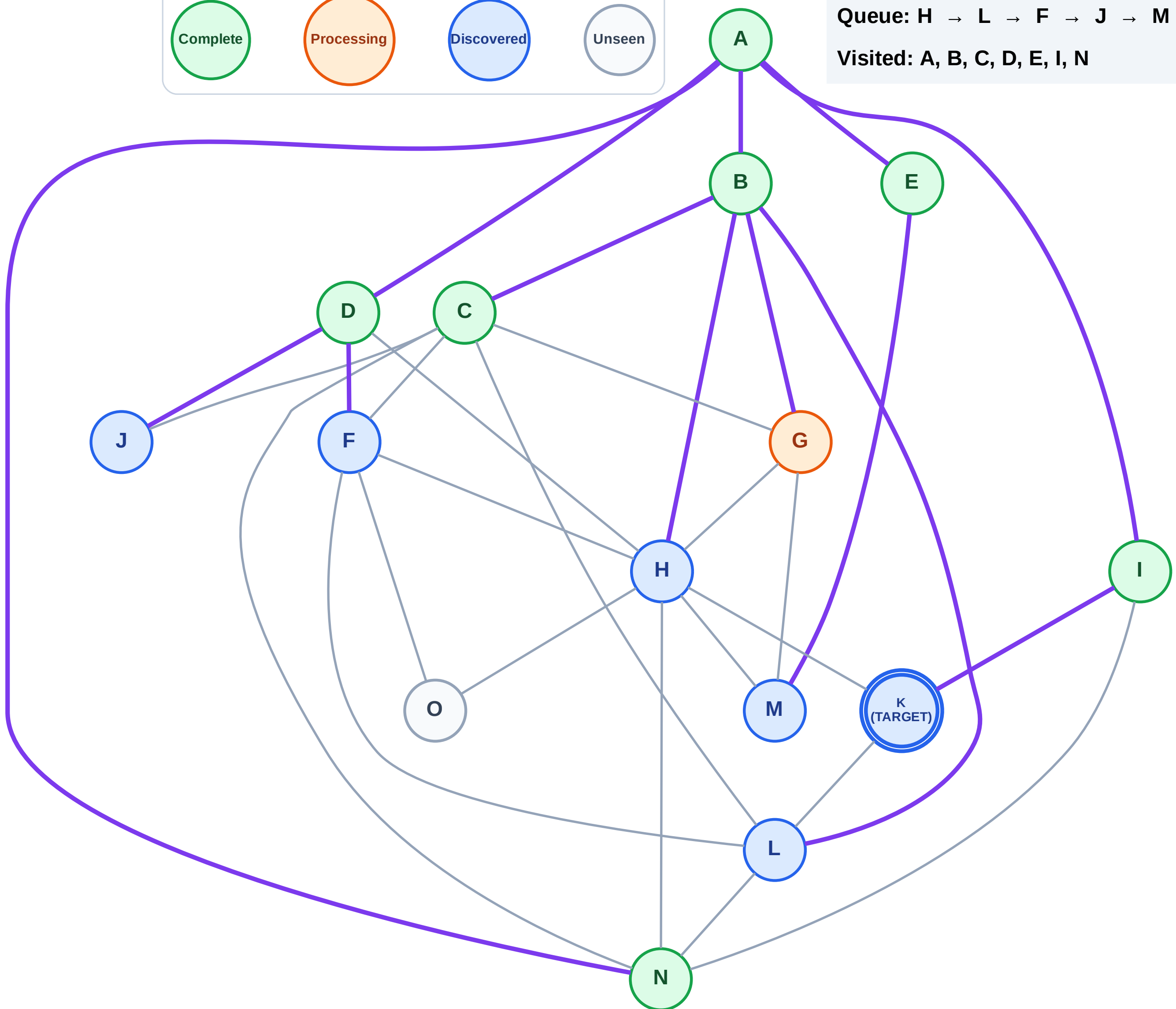
Discovered



Unseen

Queue: H → L → F → J → M → K

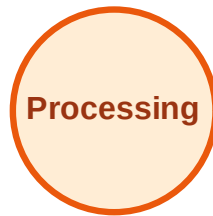
Visited: A, B, C, D, E, I, N



BFS Traversal

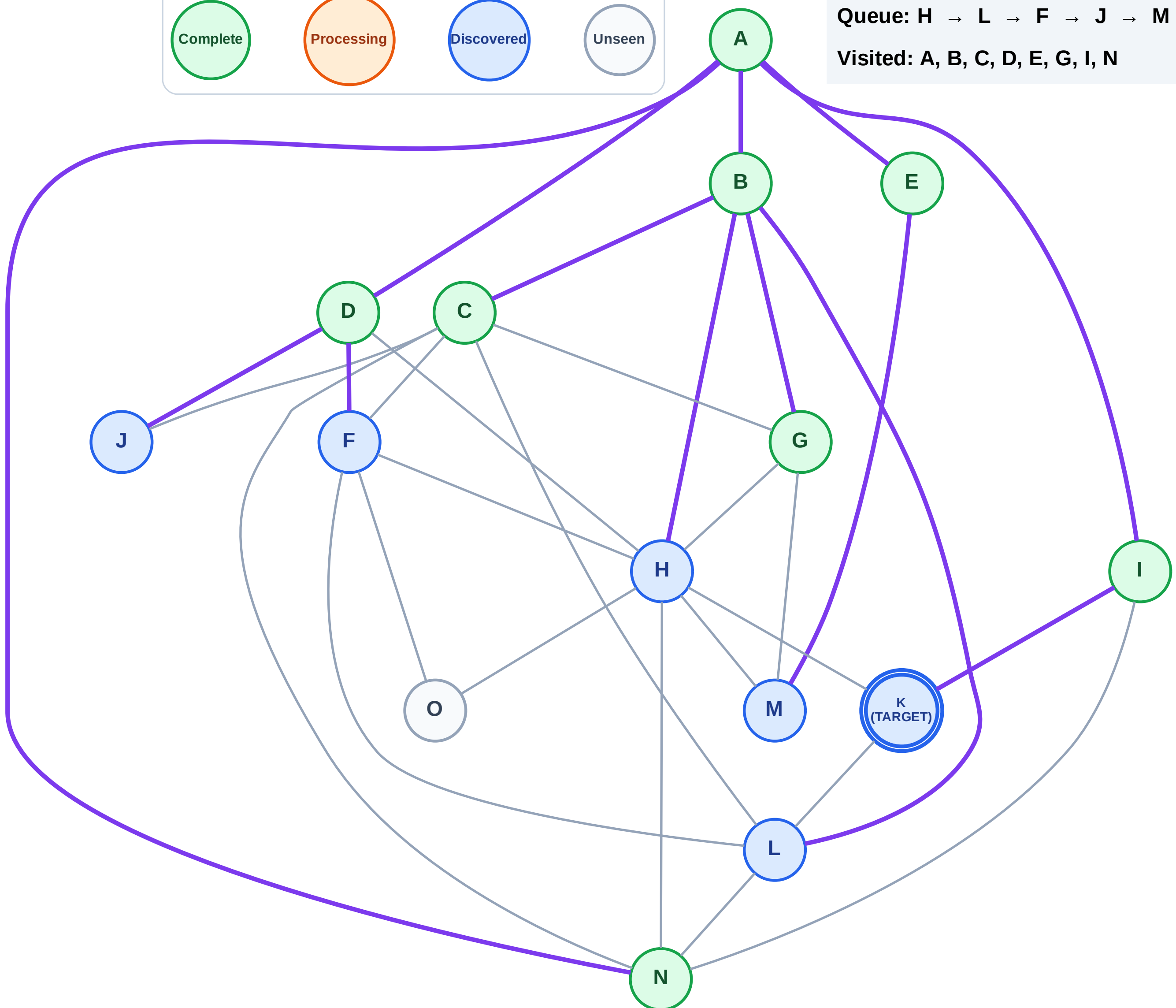
Step 17: No new neighbors from G

Legend



Queue: H → L → F → J → M → K

Visited: A, B, C, D, E, G, I, N



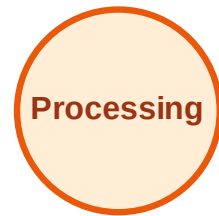
BFS Traversal

Step 18: Process node H

Legend



Complete



Processing



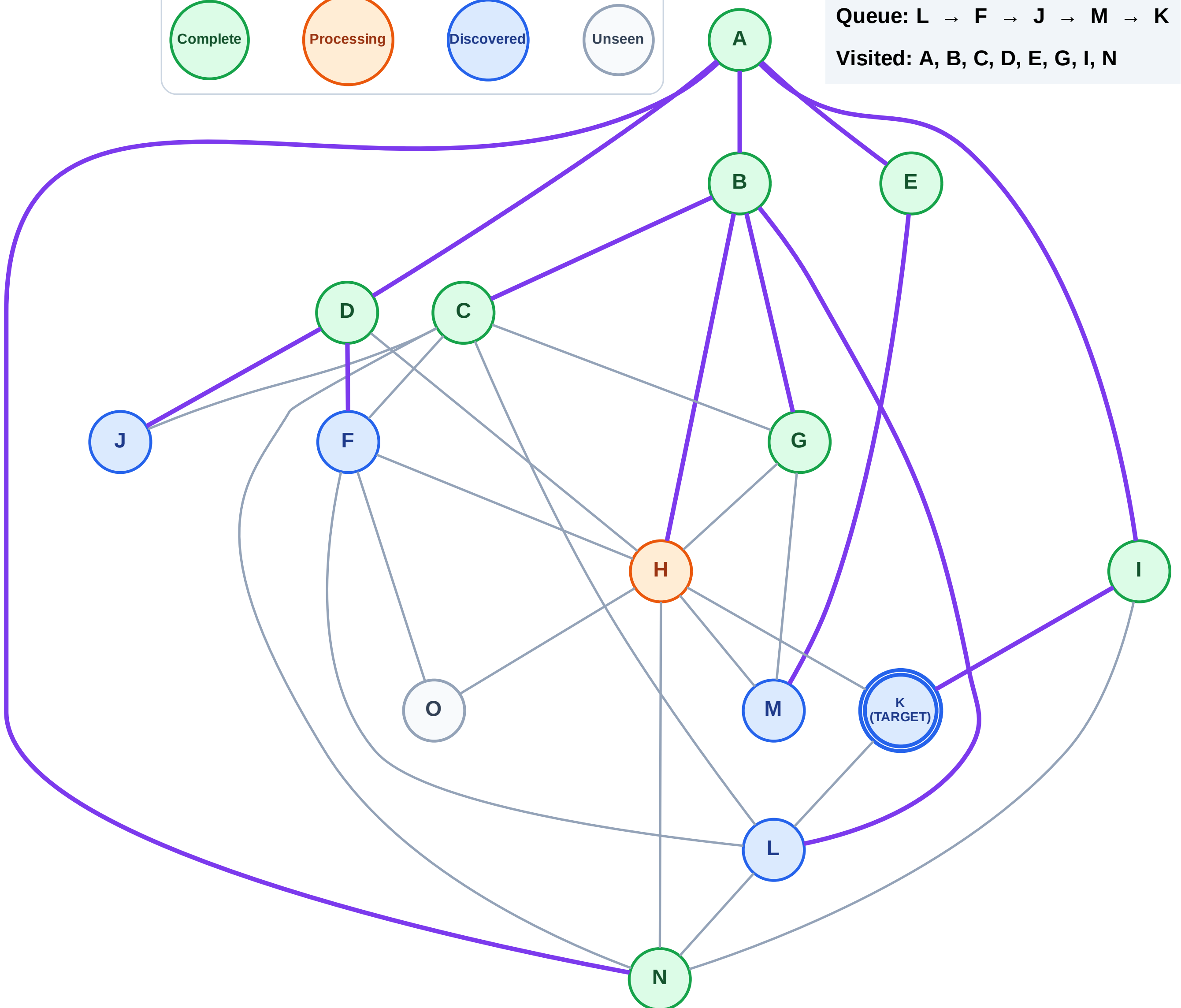
Discovered



Unseen

Queue: L → F → J → M → K

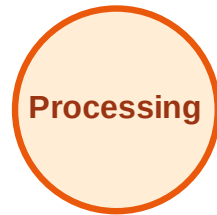
Visited: A, B, C, D, E, G, I, N



BFS Traversal

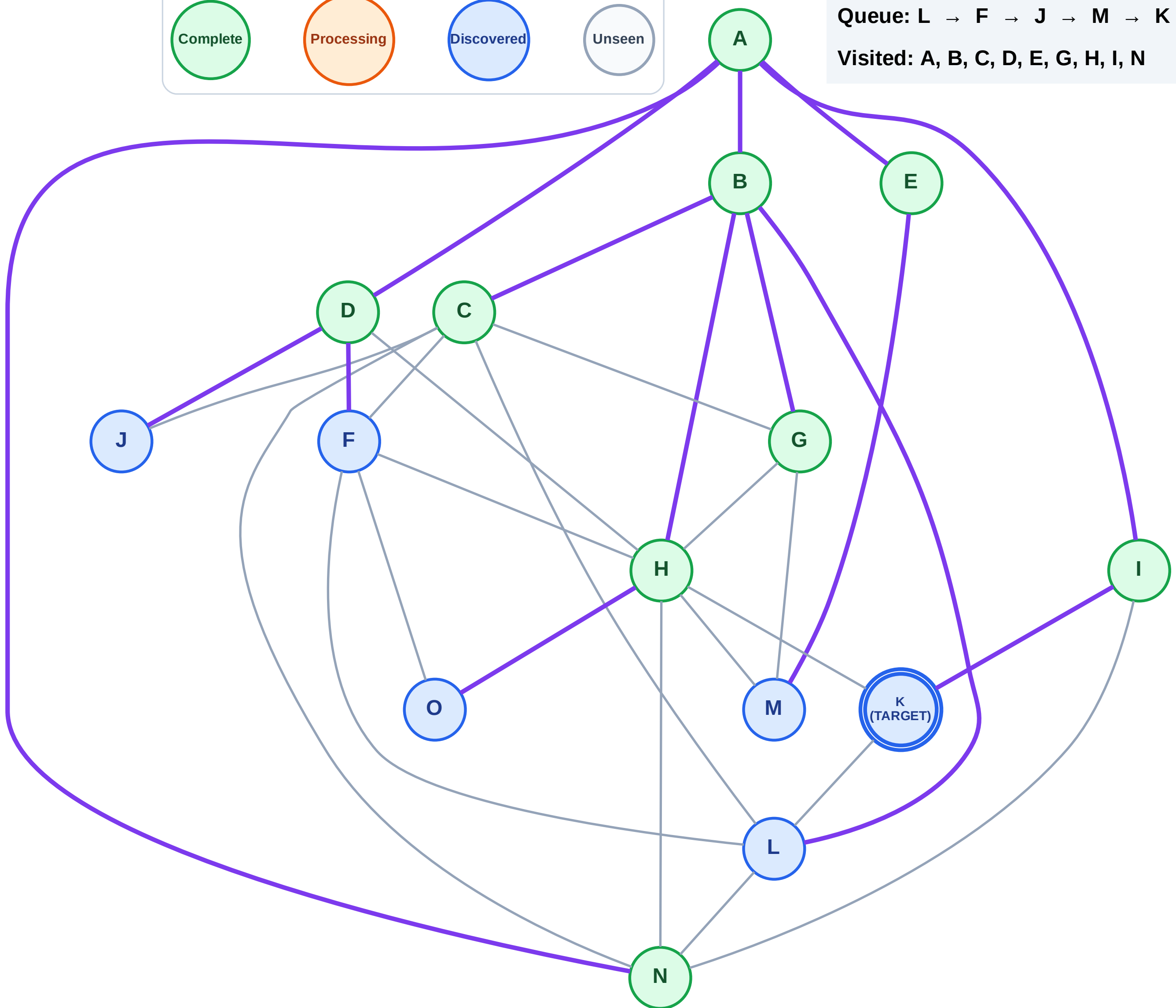
Step 19: Discover O from H

Legend



Queue: L → F → J → M → K → O

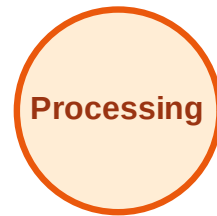
Visited: A, B, C, D, E, G, H, I, N



BFS Traversal

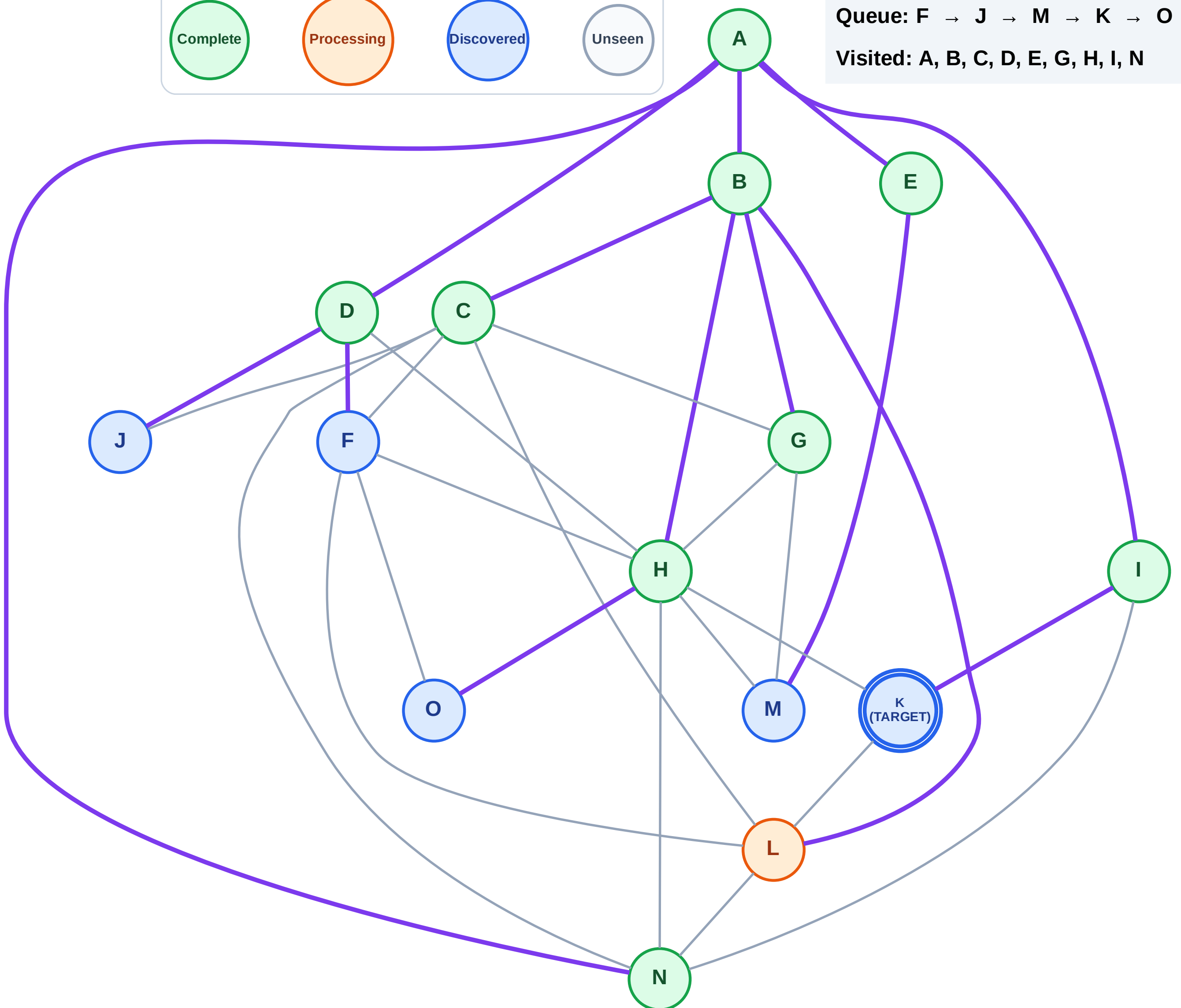
Step 20: Process node L

Legend



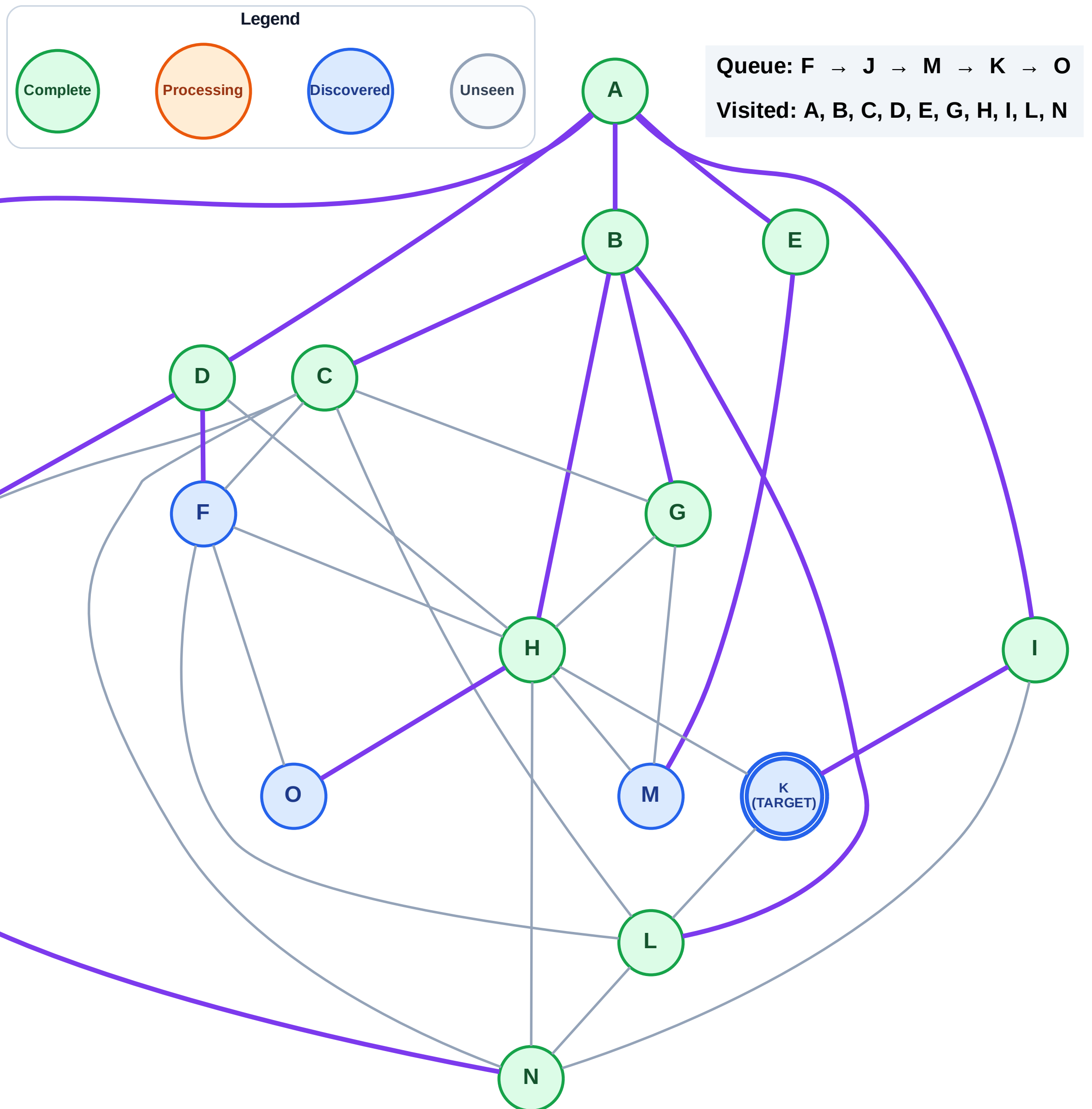
Queue: F → J → M → K → O

Visited: A, B, C, D, E, G, H, I, N

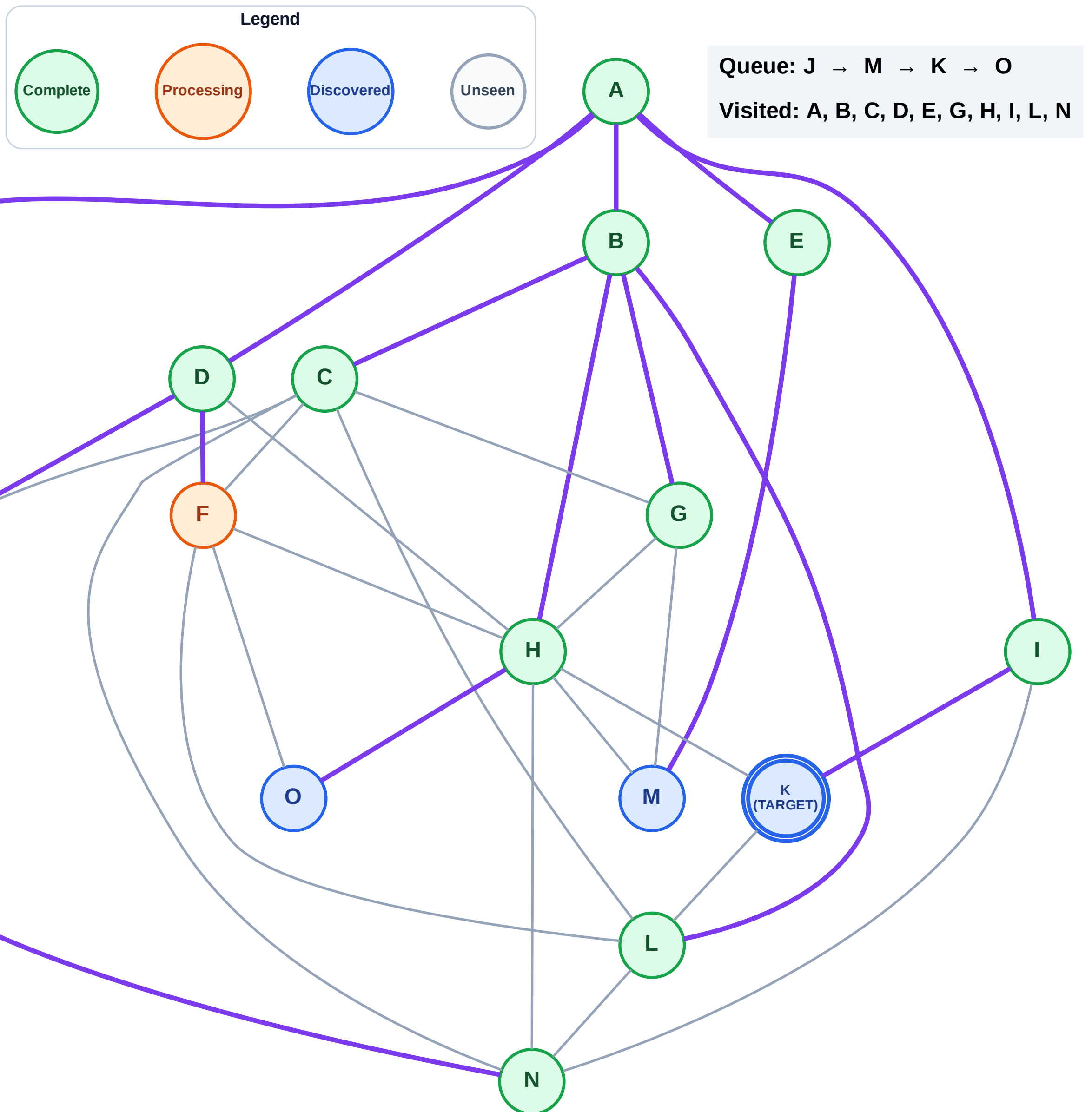


Step 21: No new neighbors from L

Step 21: No new neighbors from L



Step 22: Process node F



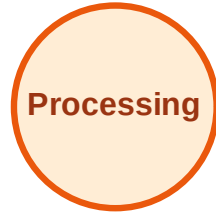
BFS Traversal

Step 23: No new neighbors from F

Legend



Complete



Processing



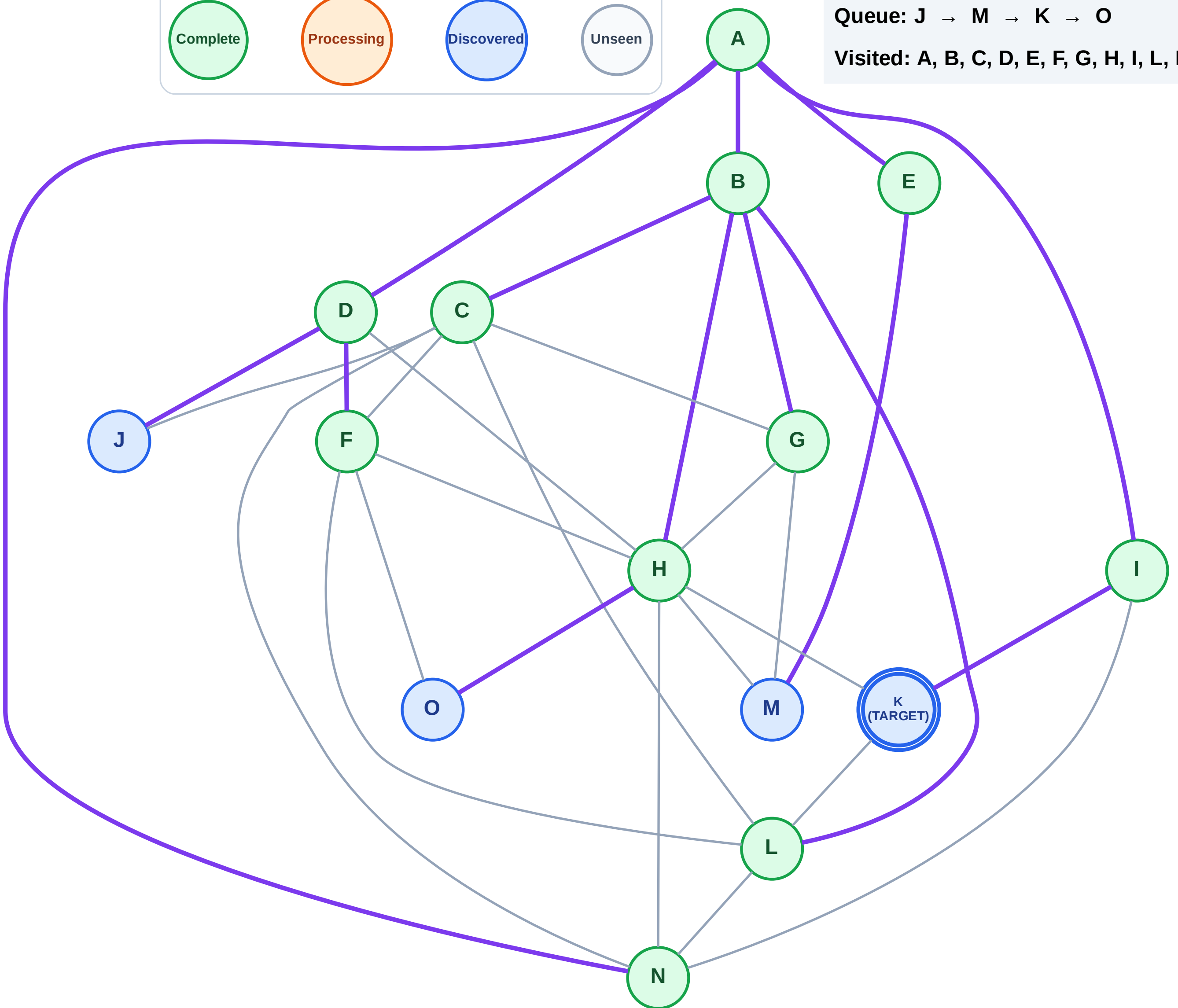
Discovered



Unseen

Queue: J → M → K → O

Visited: A, B, C, D, E, F, G, H, I, L, N



BFS Traversal

Step 24: Process node J

Legend

Complete

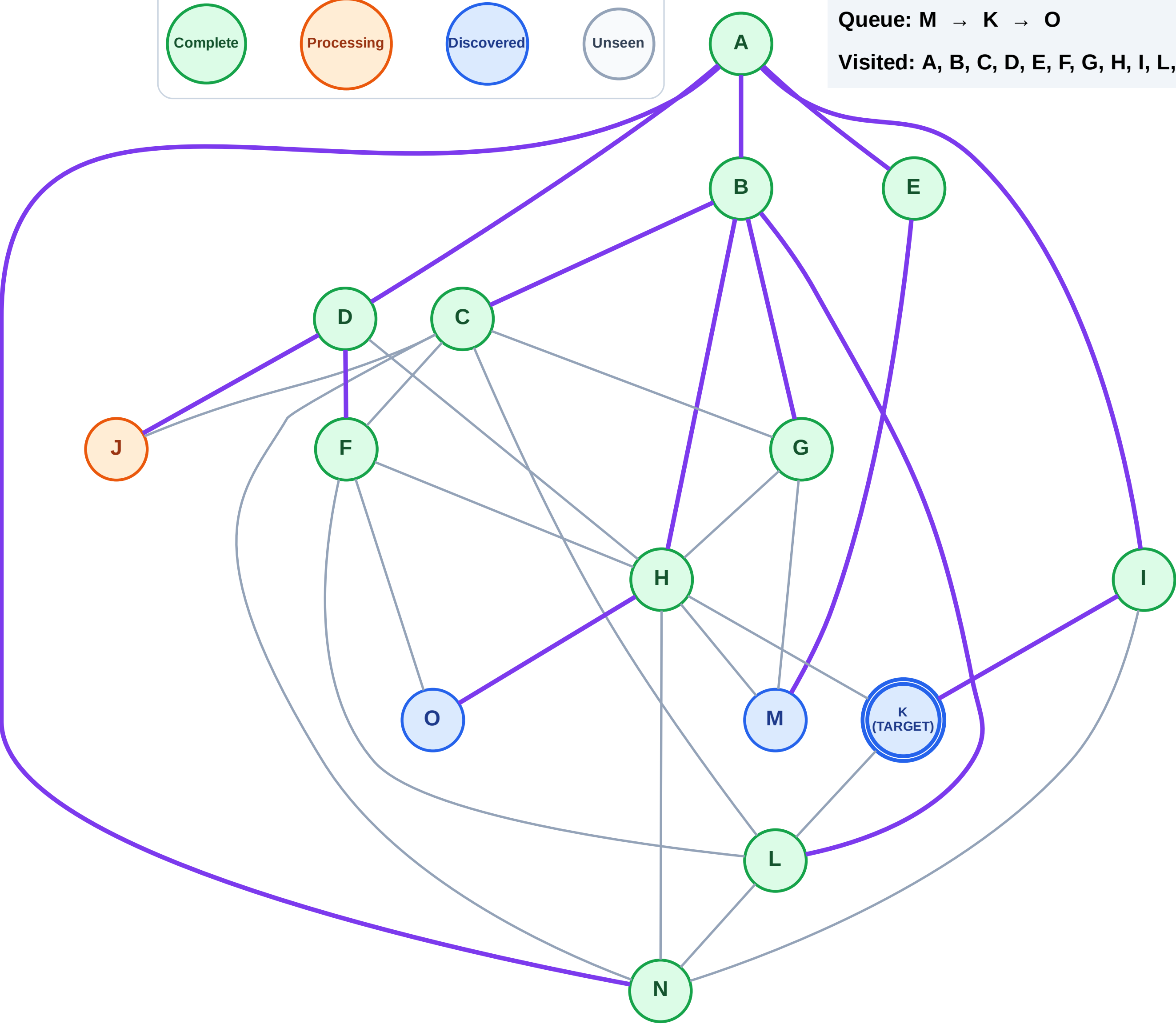
Processing

Discovered

Unseen

Queue: M → K → O

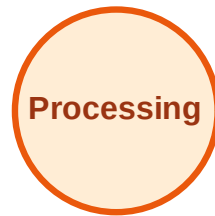
Visited: A, B, C, D, E, F, G, H, I, L, N



BFS Traversal

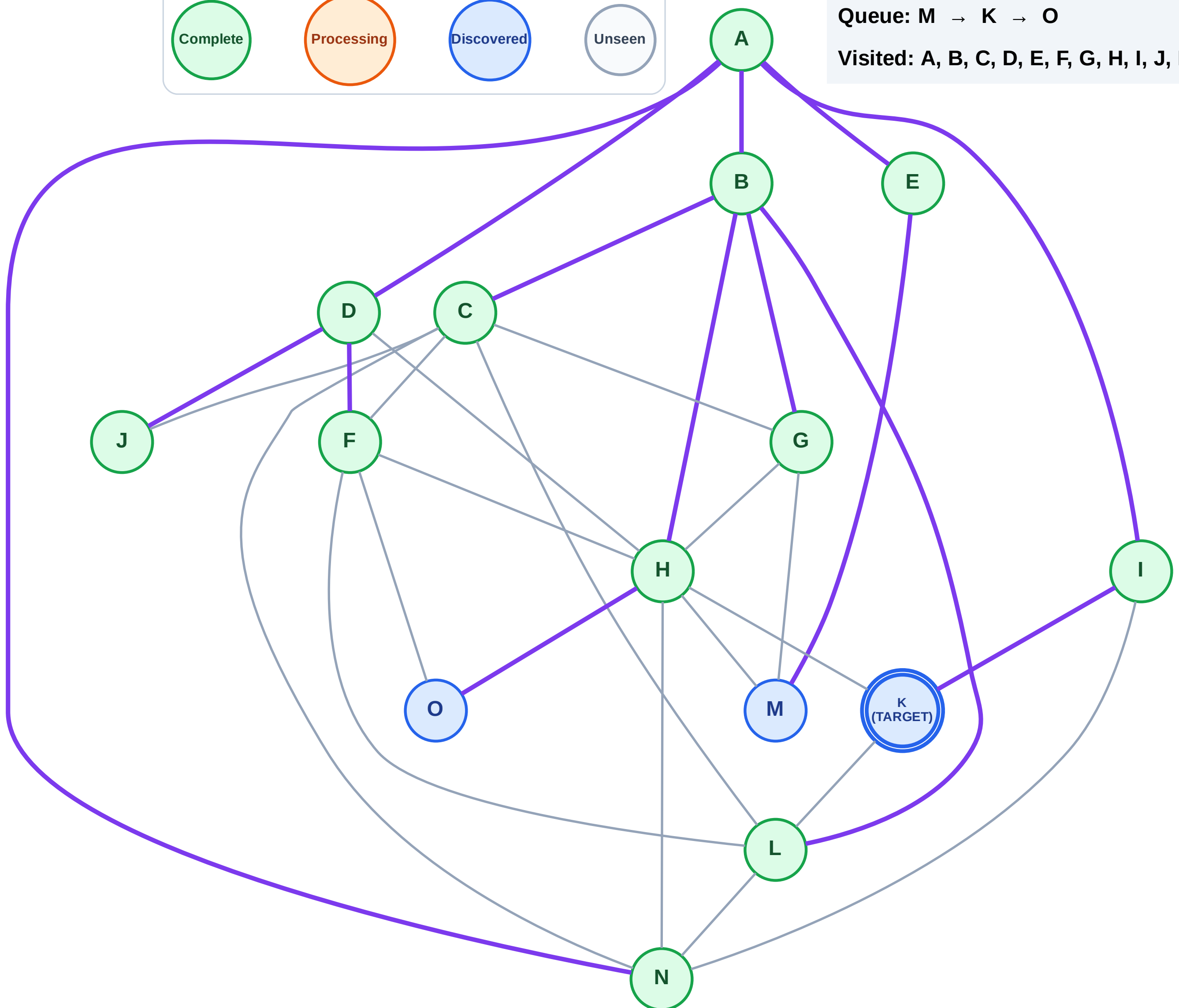
Step 25: No new neighbors from J

Legend



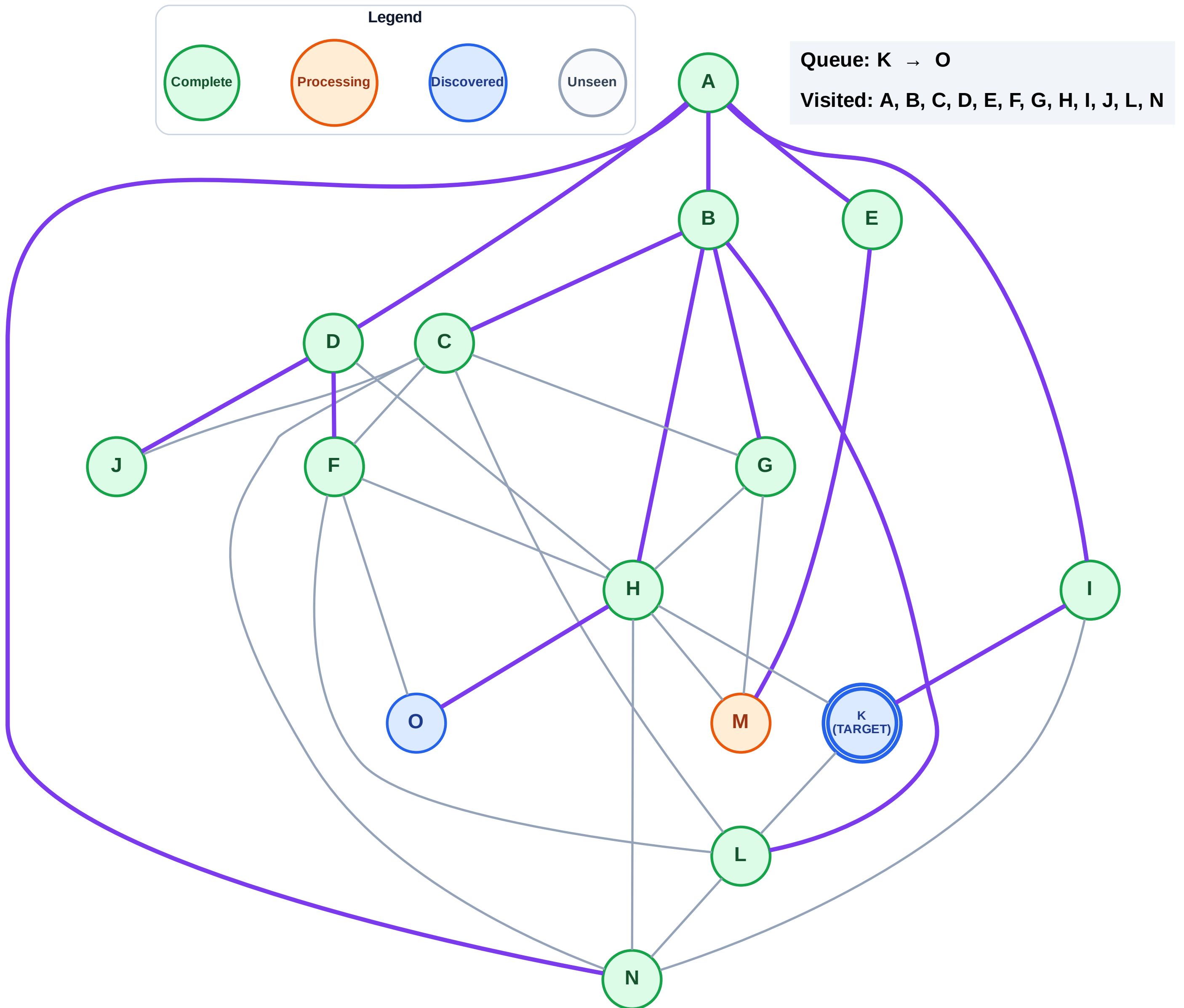
Queue: M → K → O

Visited: A, B, C, D, E, F, G, H, I, J, L, N



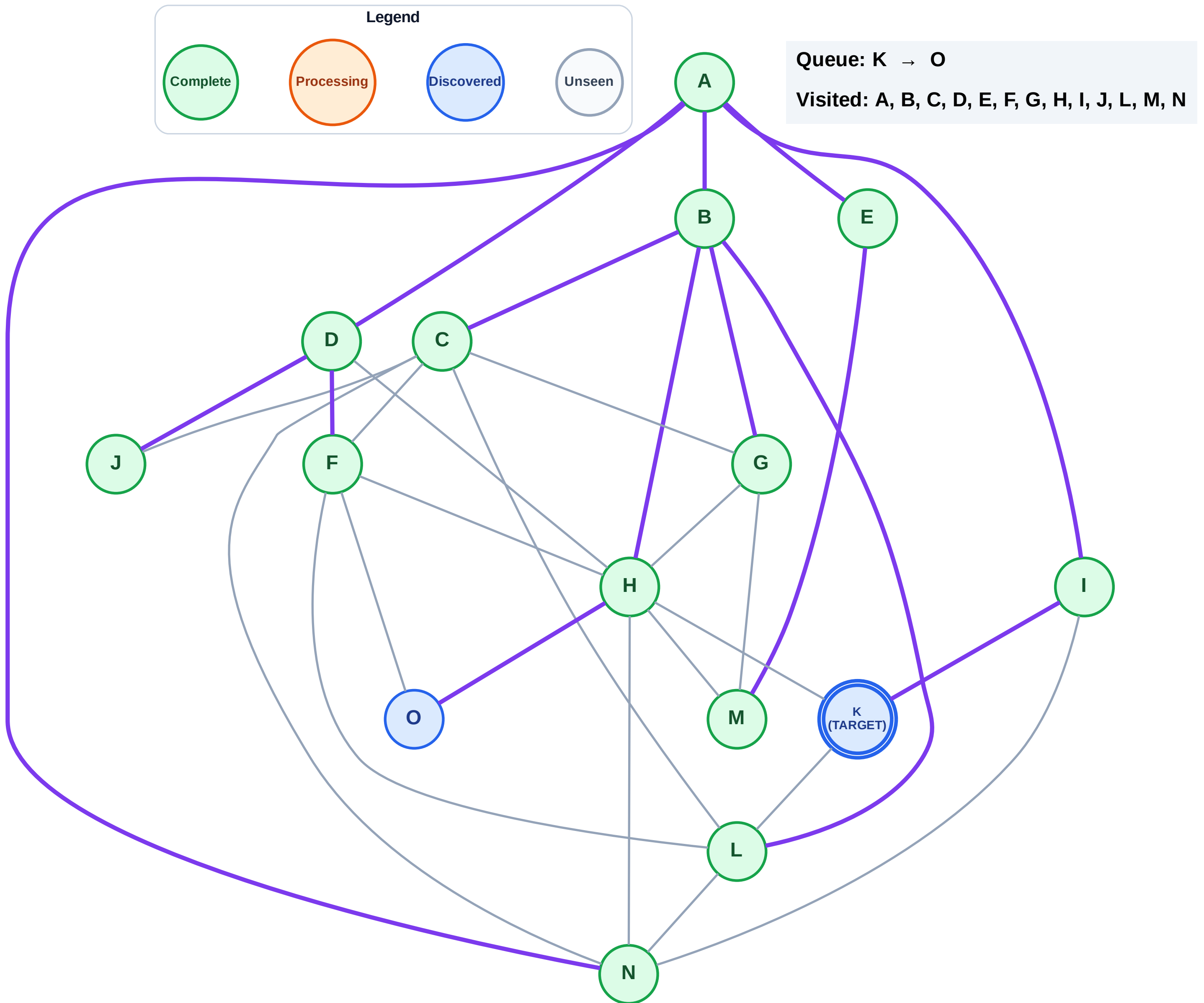
Step 26: Process node M

Step 26: Process node M



Step 27: No new neighbors from M

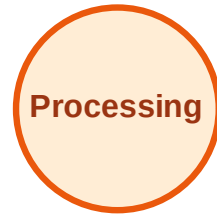
Step 27: No new neighbors from M



BFS Traversal

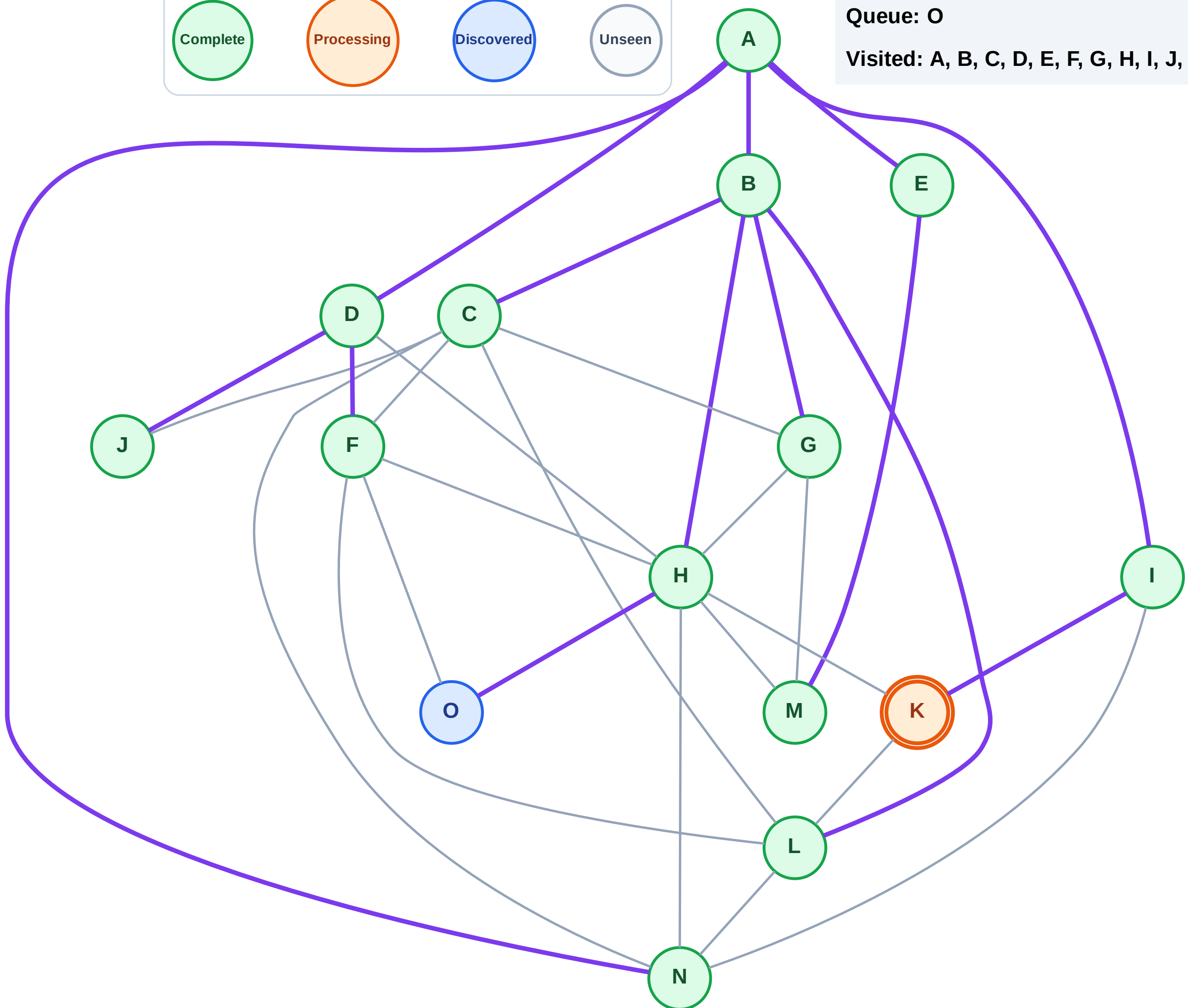
Step 28: Process node K

Legend



Queue: O

Visited: A, B, C, D, E, F, G, H, I, J, L, M, N



BFS Traversal

Step 29: TARGET NODE K REACHED! Path (2 hops): A → I → K

Legend

Complete

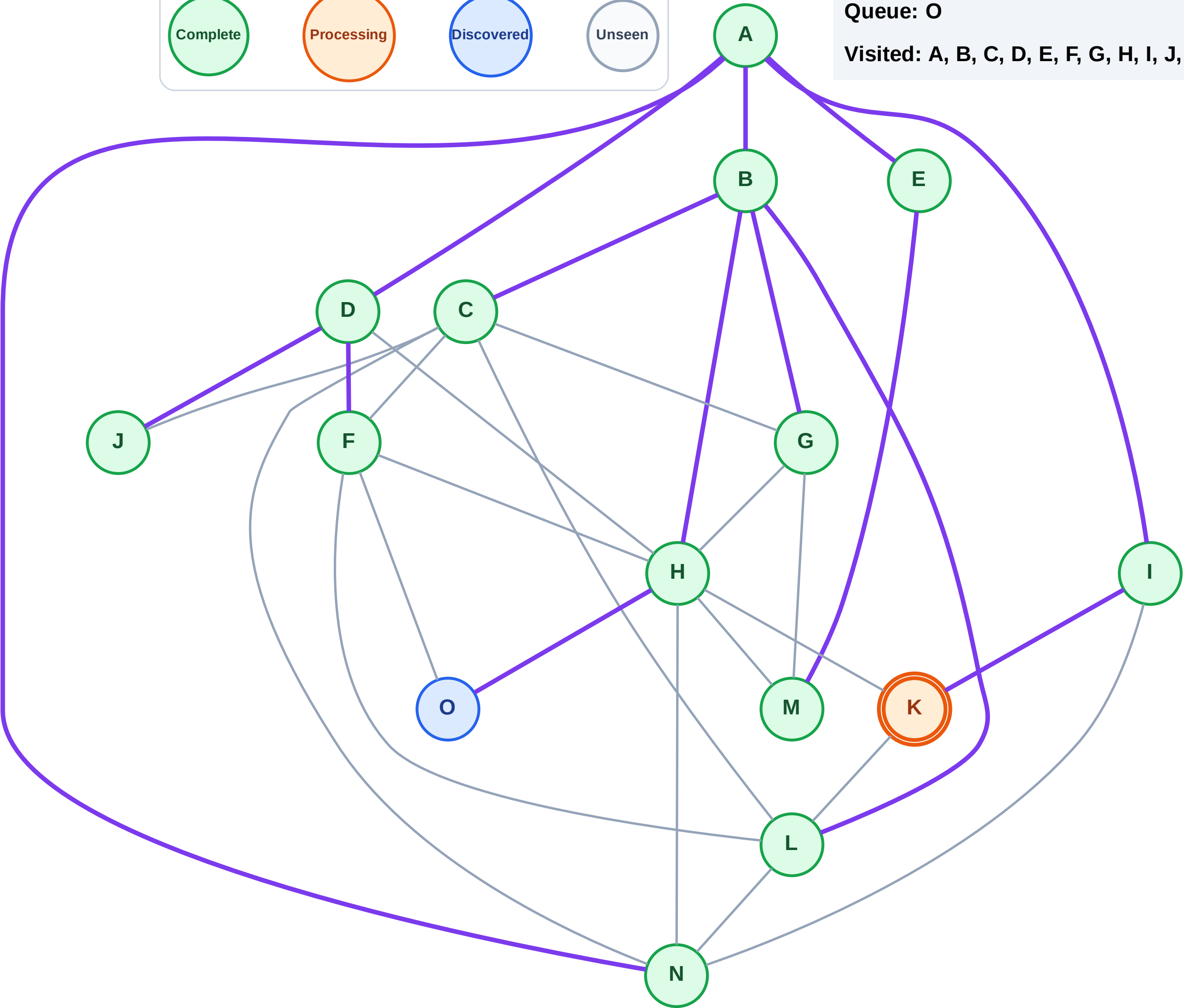
Processing

Discovered

Unseen

Queue: O

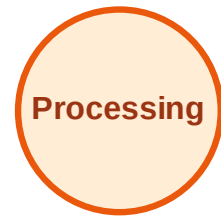
Visited: A, B, C, D, E, F, G, H, I, J, K, L, M, N



DFS Traversal

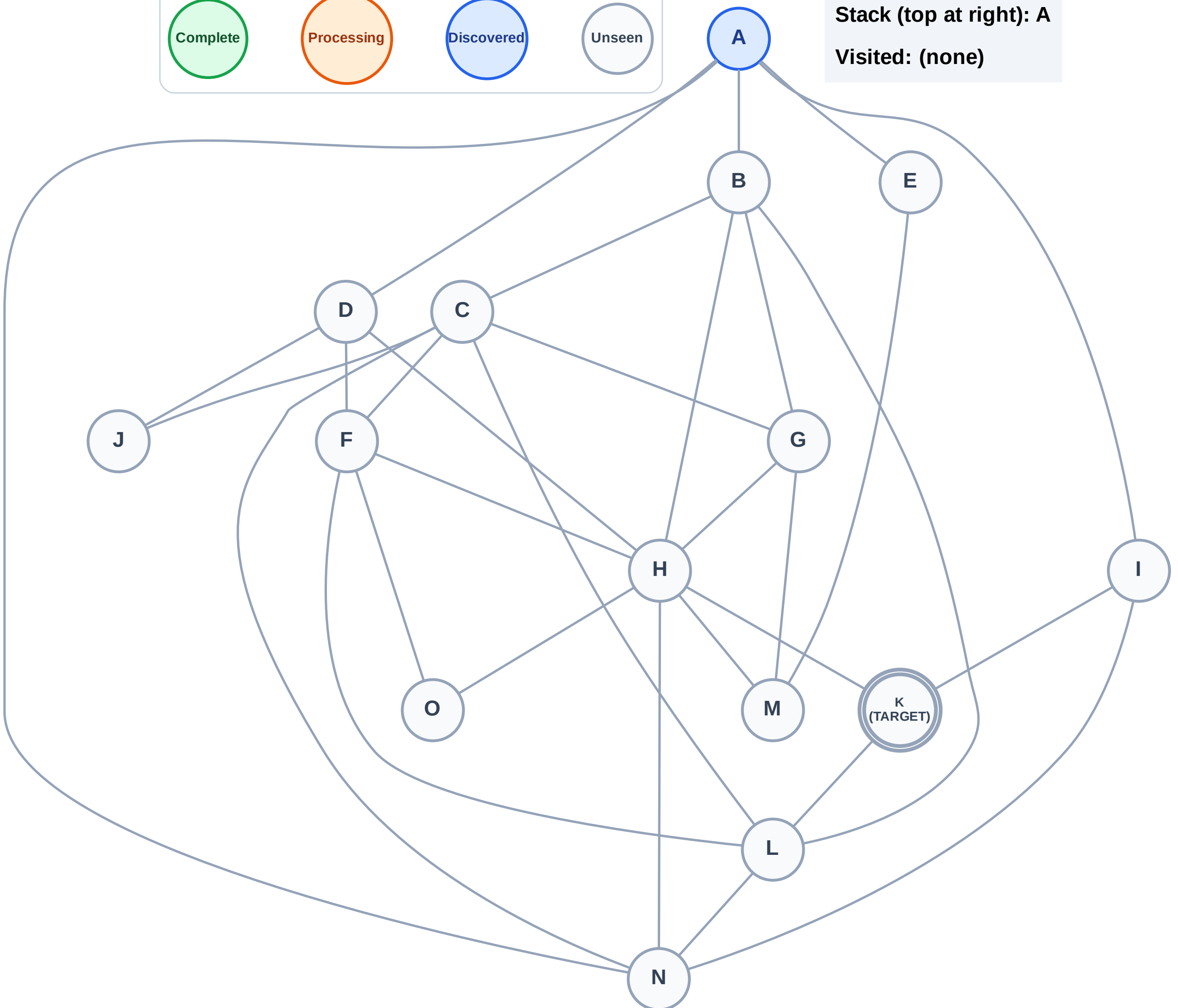
Step 1: Start at A (Seeking target node K)

Legend



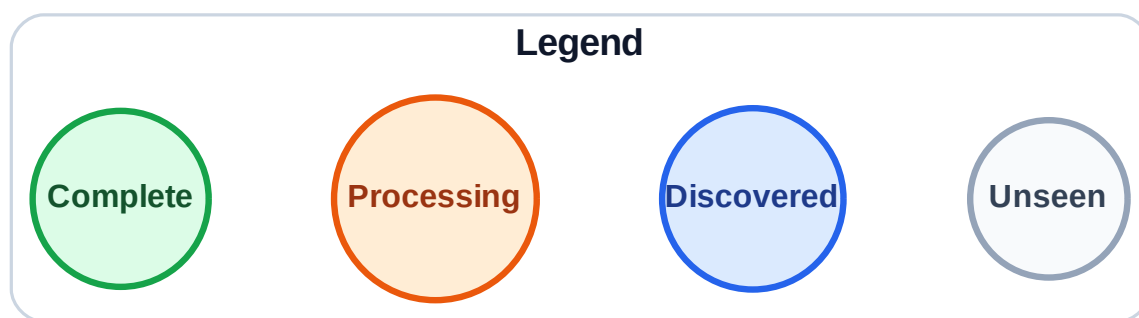
Stack (top at right): A

Visited: (none)



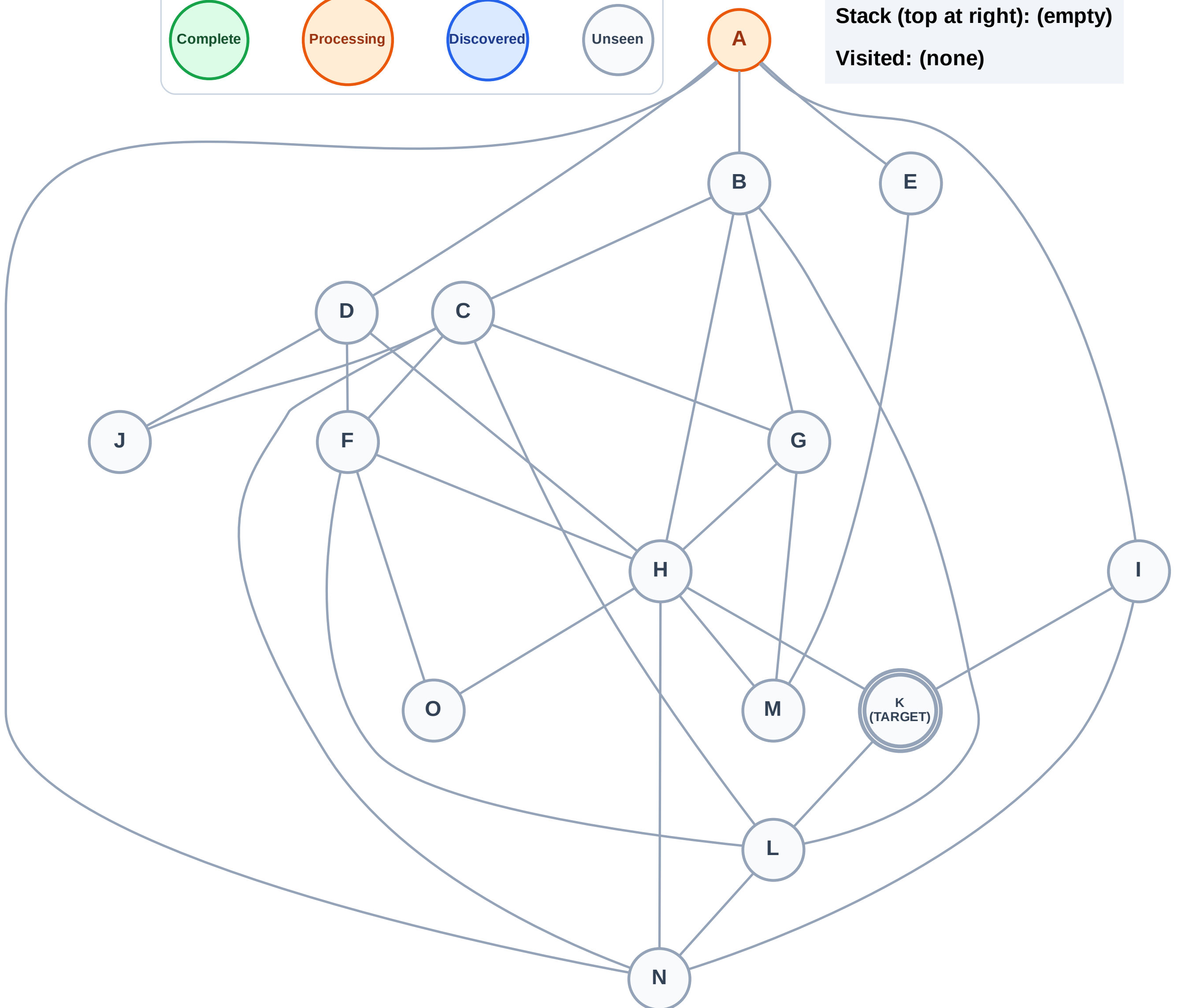
DFS Traversal

Step 2: Process node A



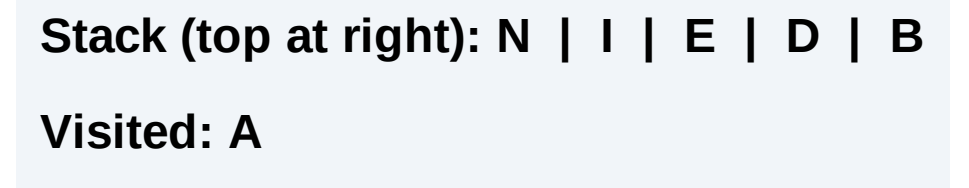
Stack (top at right): (empty)

Visited: (none)



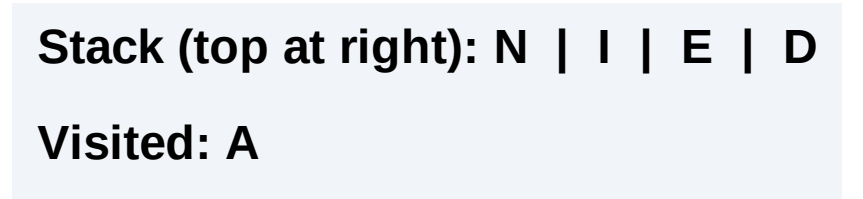
Step 3: Discover N, I, E, D, B from A

Step 3: Discover N, I, E, D, B from A



Step 4: Process node B

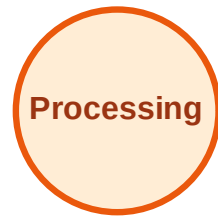
Step 4: Process node B



DFS Traversal

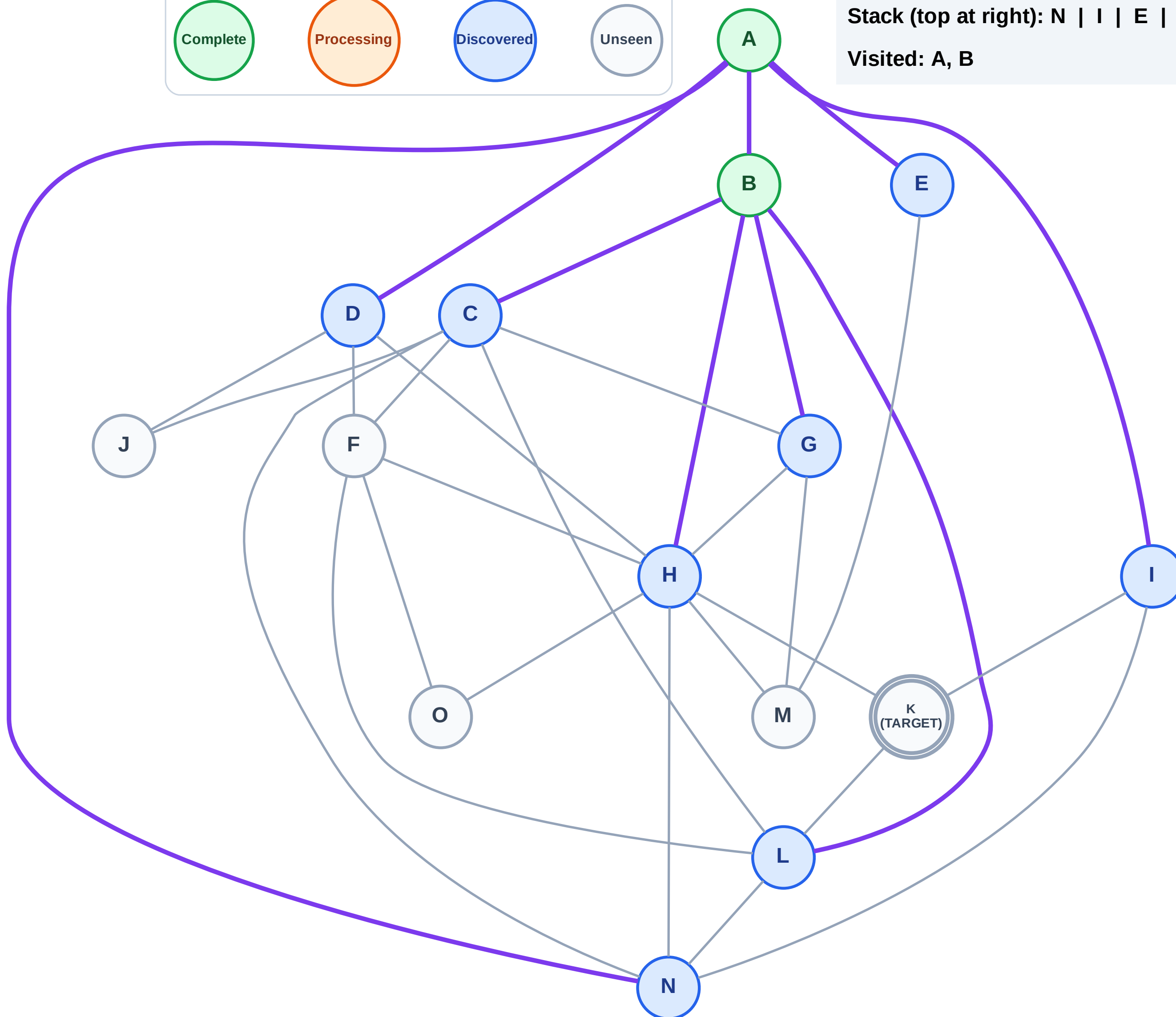
Step 5: Discover L, H, G, C from B

Legend



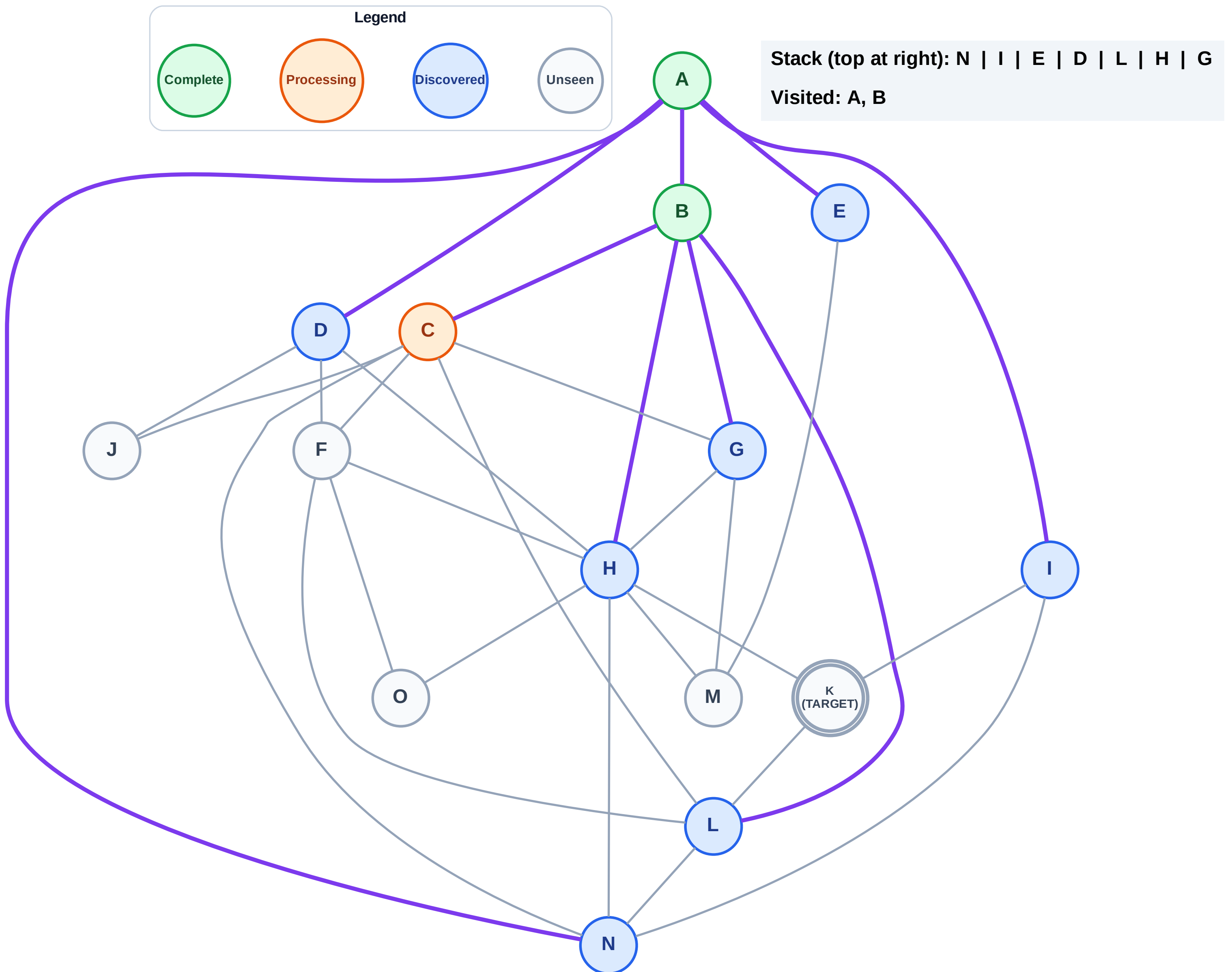
Stack (top at right): N | I | E | D | L | H | G | C

Visited: A, B



Step 6: Process node C

Step 6: Process node C



DFS Traversal

Step 7: Discover J, F from C

Legend

Complete

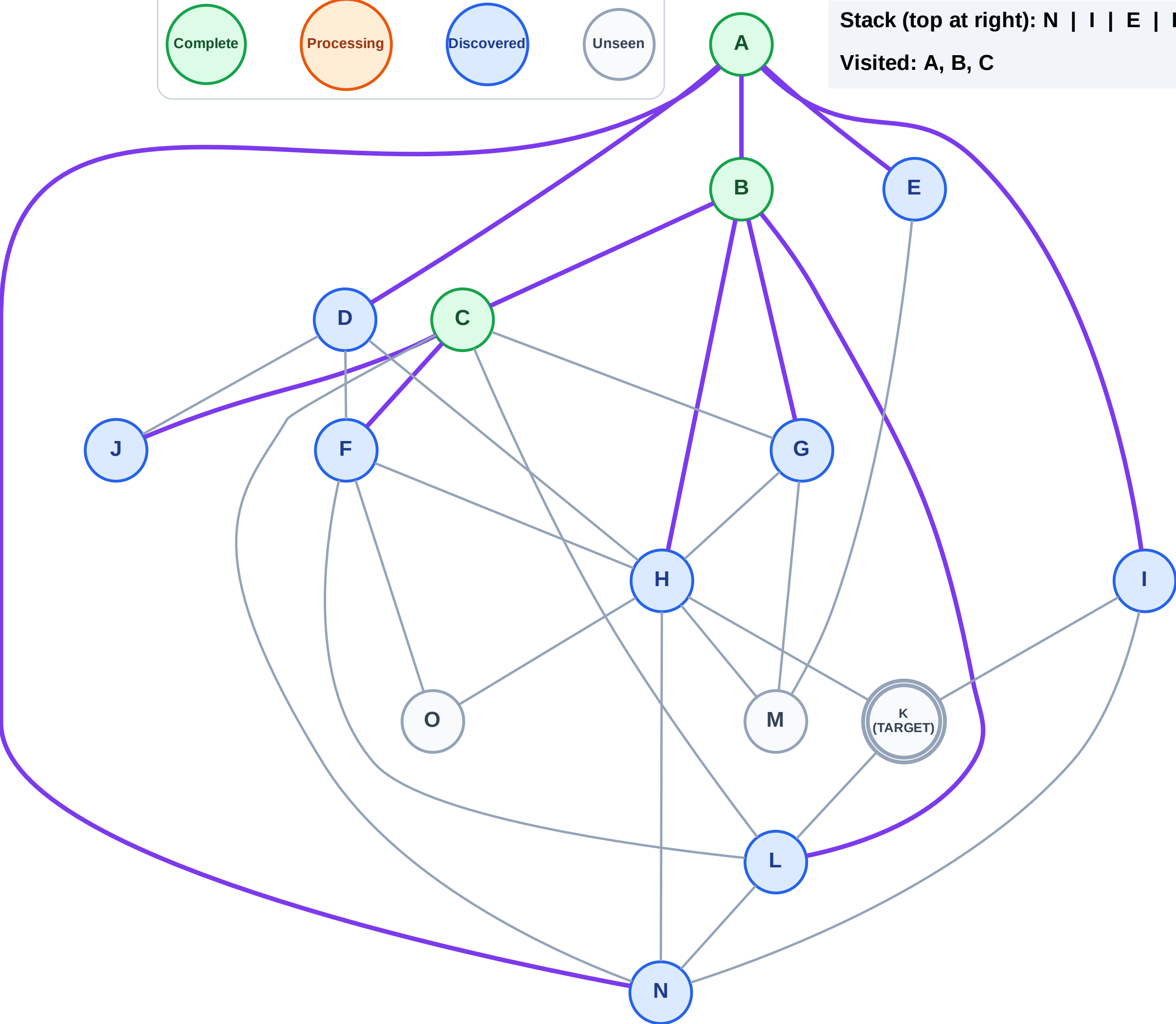
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J | F

Visited: A, B, C



DFS Traversal

Step 8: Process node F

Legend

Complete

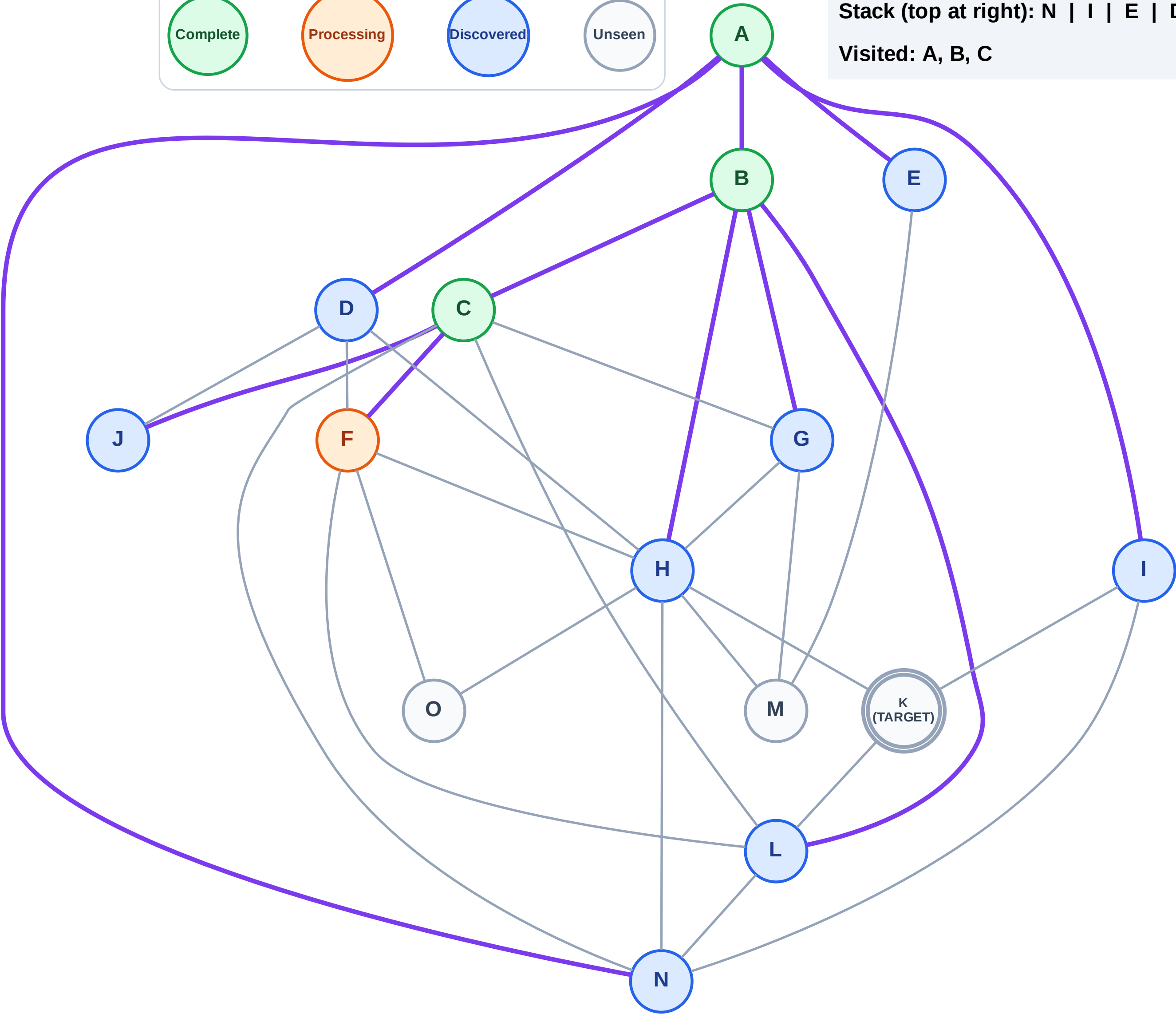
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H | G | J

Visited: A, B, C



DFS Traversal

Step 9: Discover O from F

Legend

Complete

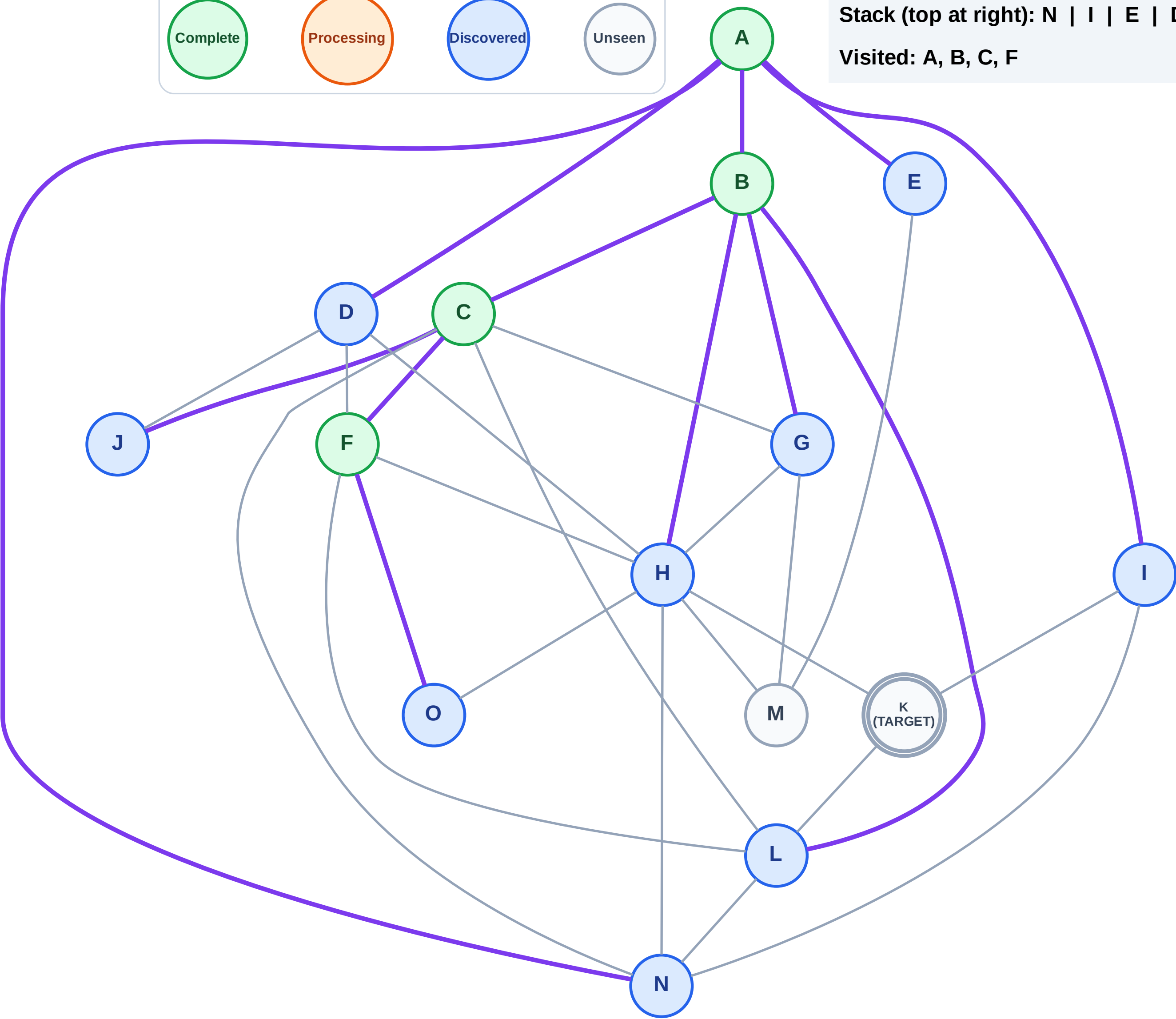
Processing

Discovered

Unseen

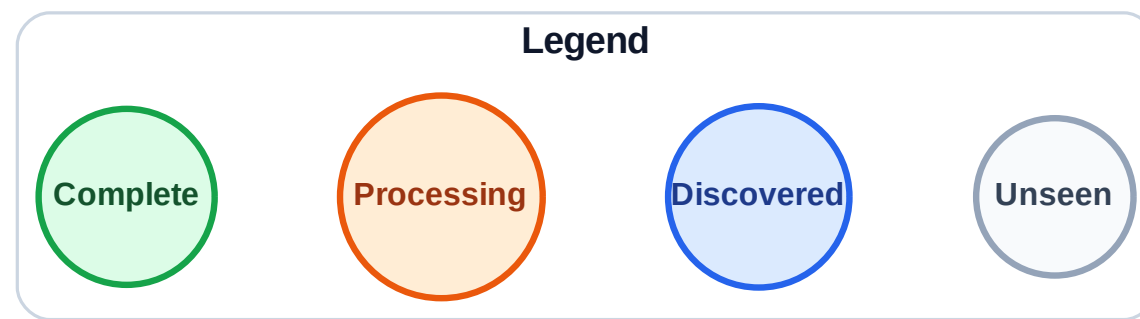
Stack (top at right): N | I | E | D | L | H | G | J | O

Visited: A, B, C, F

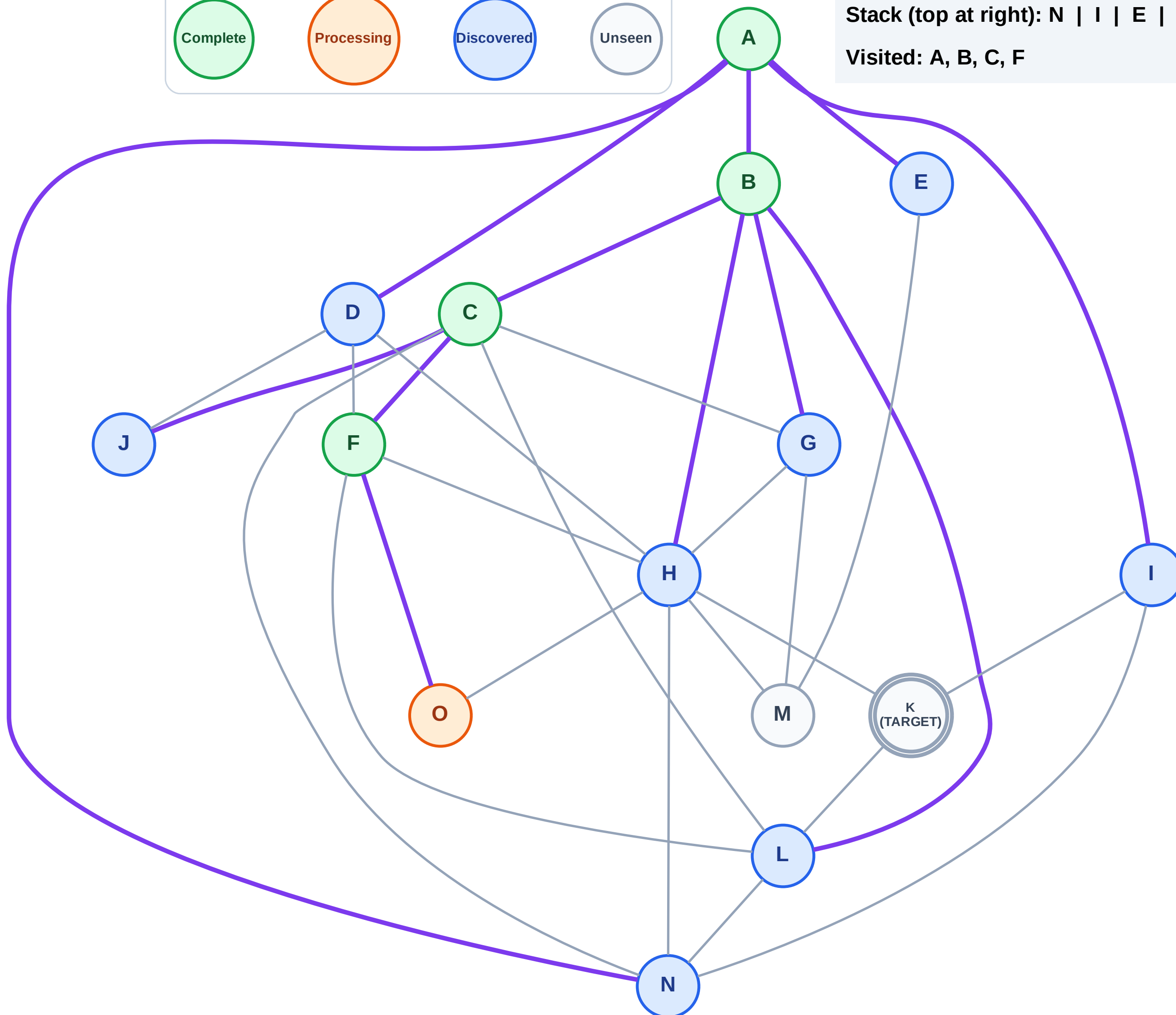


DFS Traversal

Step 10: Process node O



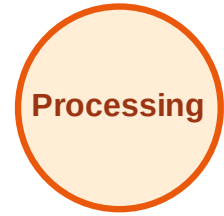
Stack (top at right): N | I | E | D | L | H | G | J
Visited: A, B, C, F



DFS Traversal

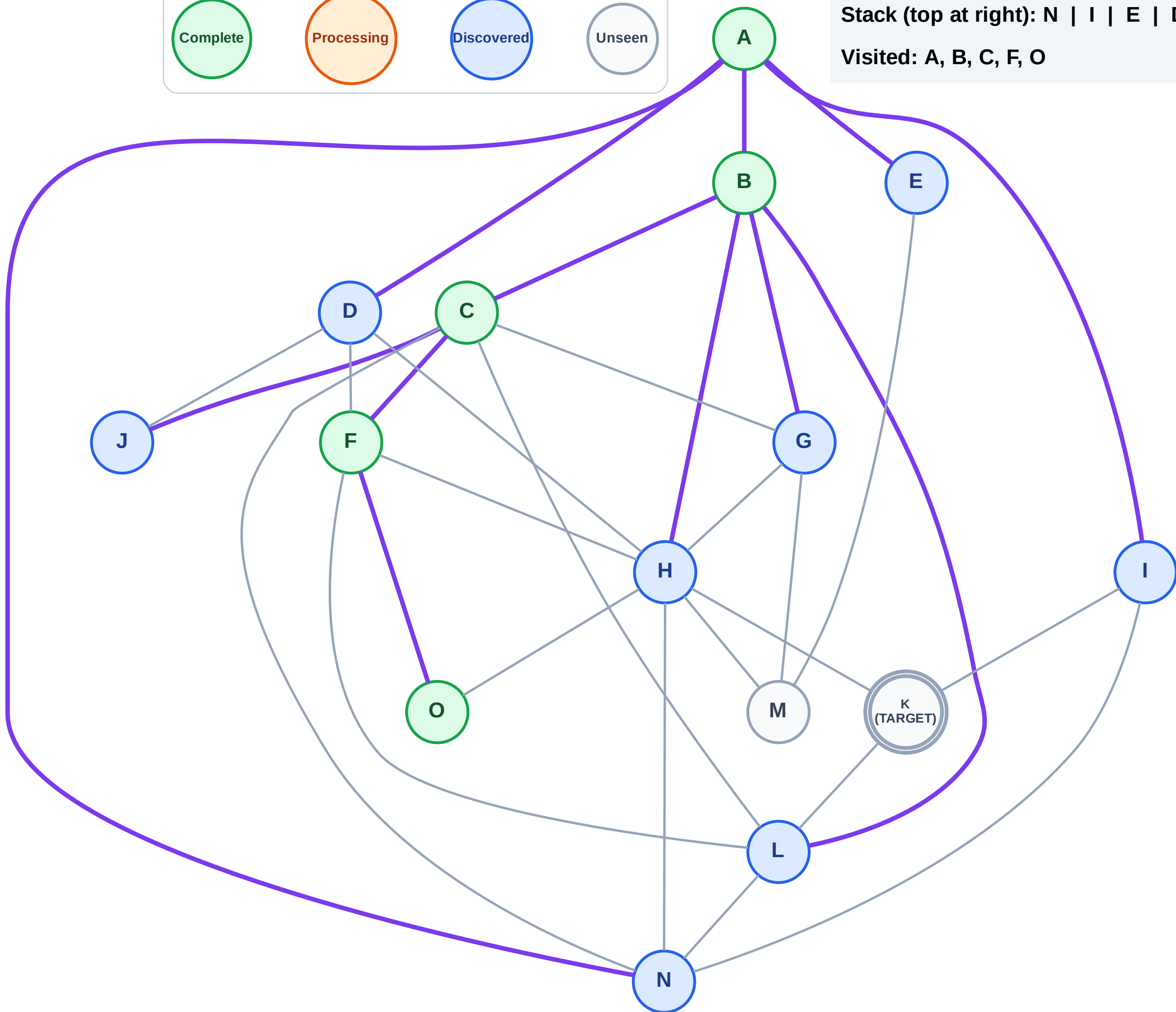
Step 11: No new neighbors from O

Legend



Stack (top at right): N | I | E | D | L | H | G | J

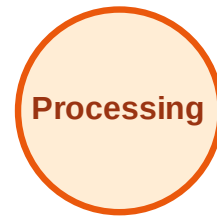
Visited: A, B, C, F, O



DFS Traversal

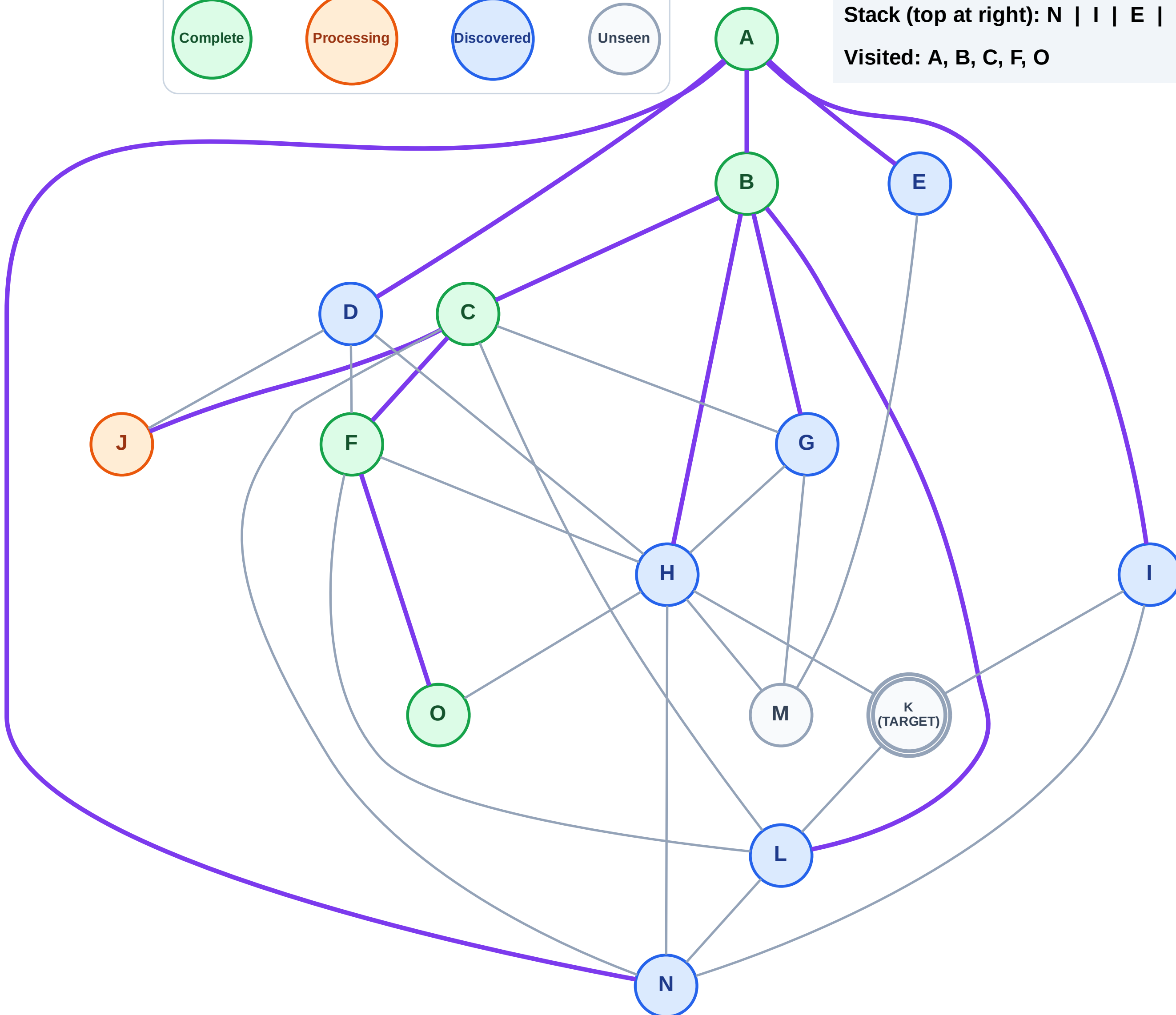
Step 12: Process node J

Legend



Stack (top at right): N | I | E | D | L | H | G

Visited: A, B, C, F, O

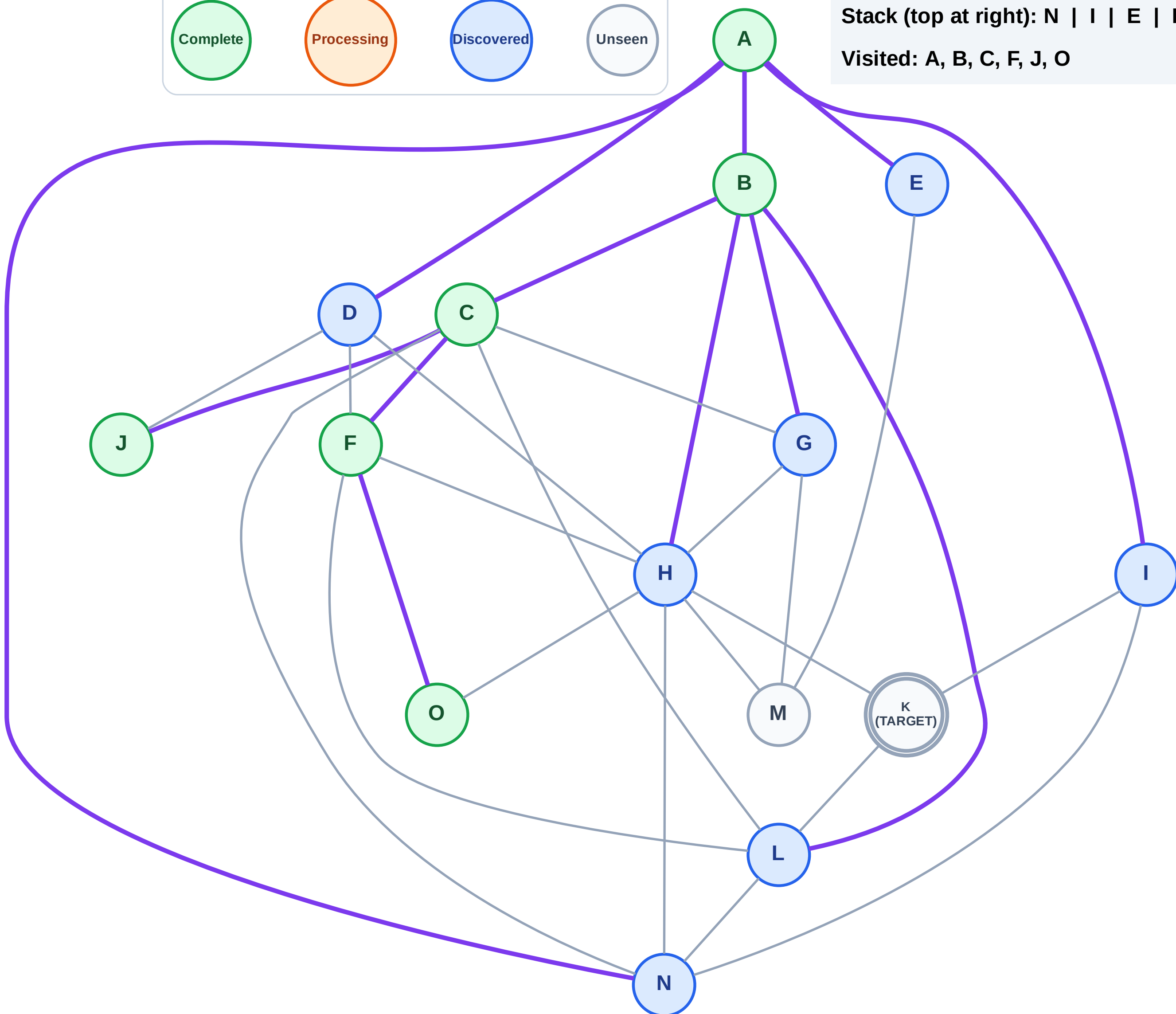


Step 13: No new neighbors from J

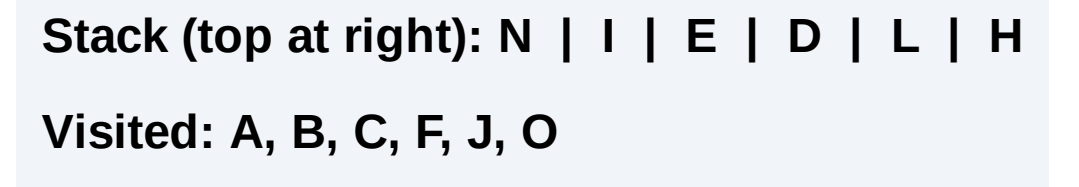
Legend

Unseen

Visited: A, B, C, F, J, O

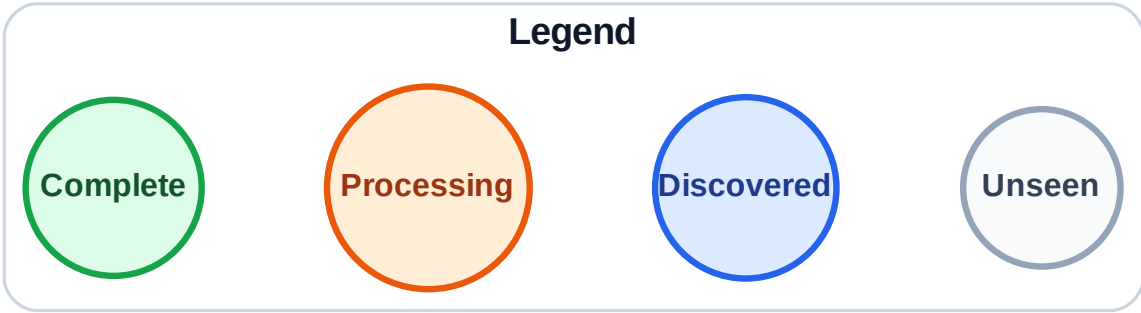


Step 14: Process node G



Step 15: Discover M from G

Step 15: Discover M from G



DFS Traversal

Step 16: Process node M

Legend

Complete

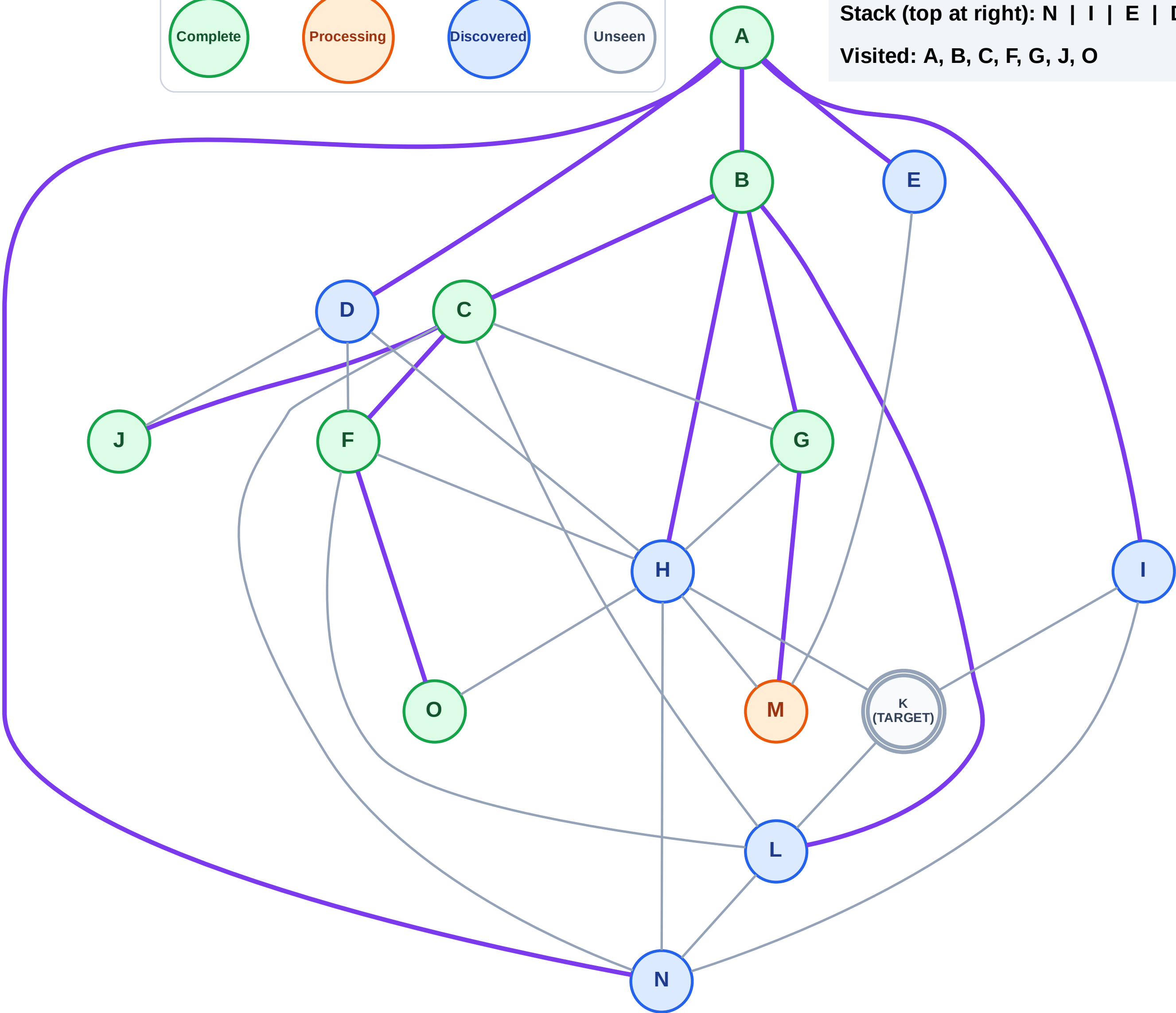
Processing

Discovered

Unseen

Stack (top at right): N | I | E | D | L | H

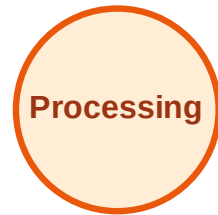
Visited: A, B, C, F, G, J, O



DFS Traversal

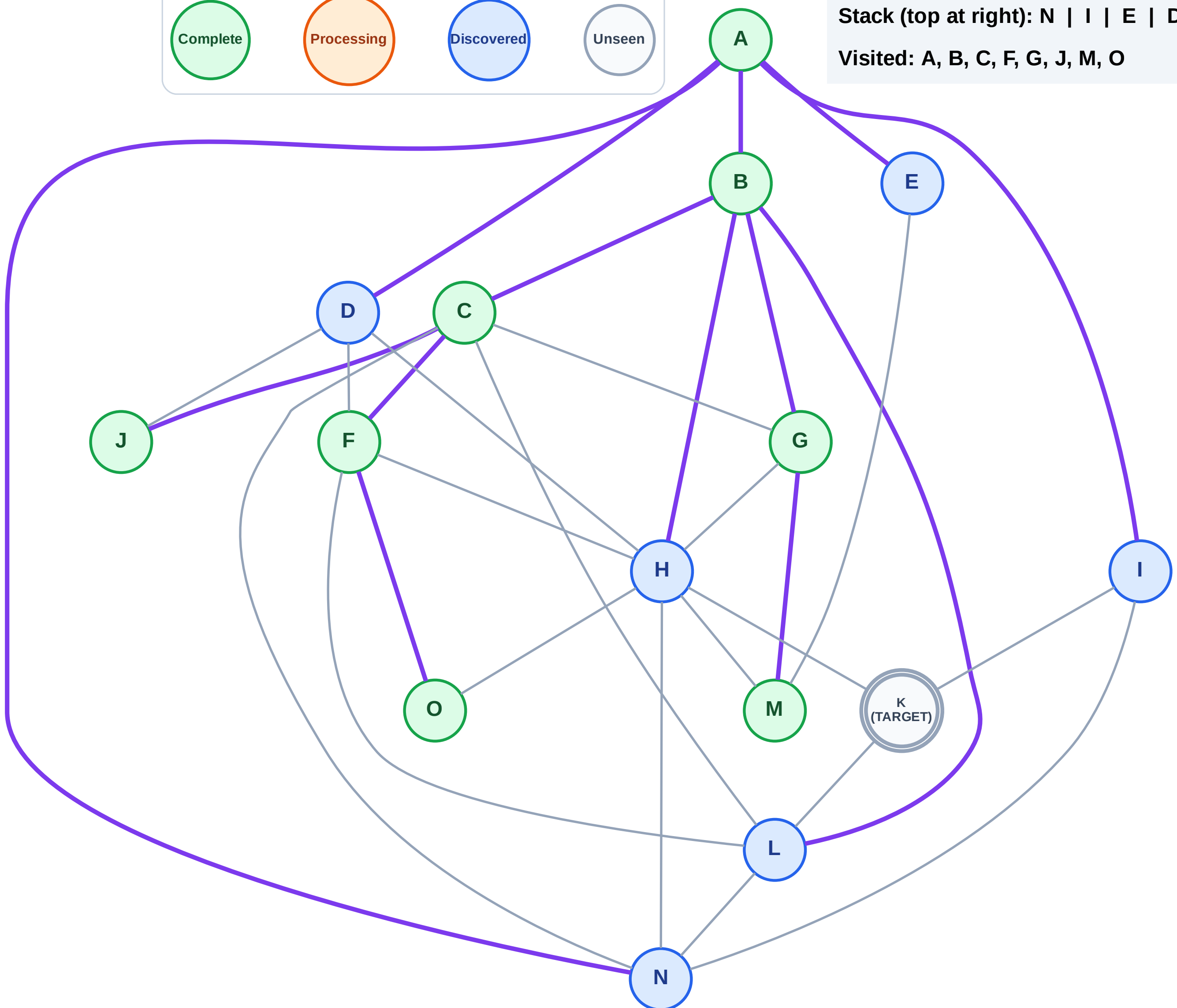
Step 17: No new neighbors from M

Legend



Stack (top at right): N | I | E | D | L | H

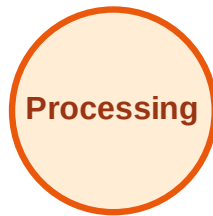
Visited: A, B, C, F, G, J, M, O



DFS Traversal

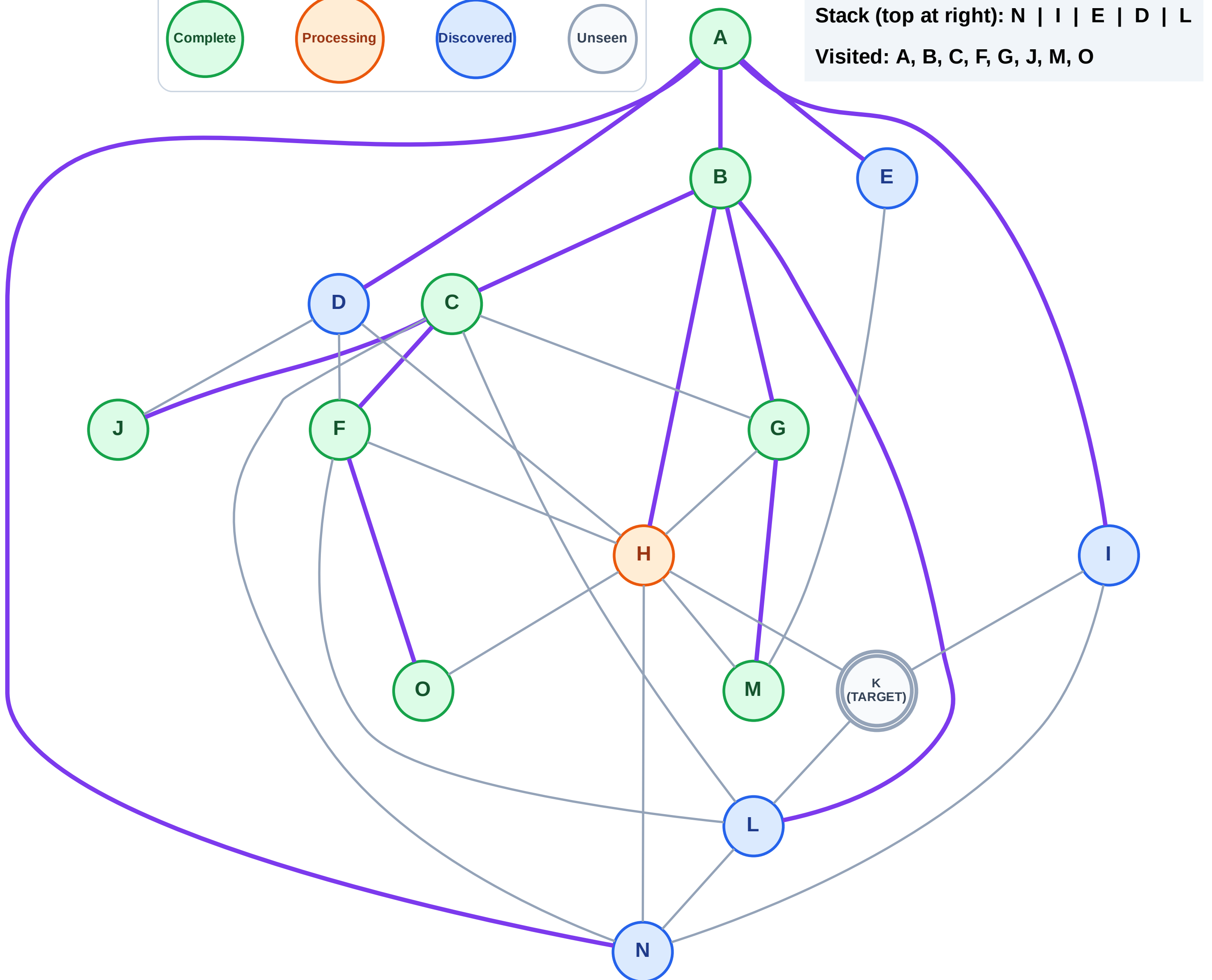
Step 18: Process node H

Legend



Stack (top at right): N | I | E | D | L

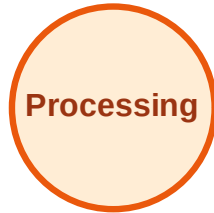
Visited: A, B, C, F, G, J, M, O



DFS Traversal

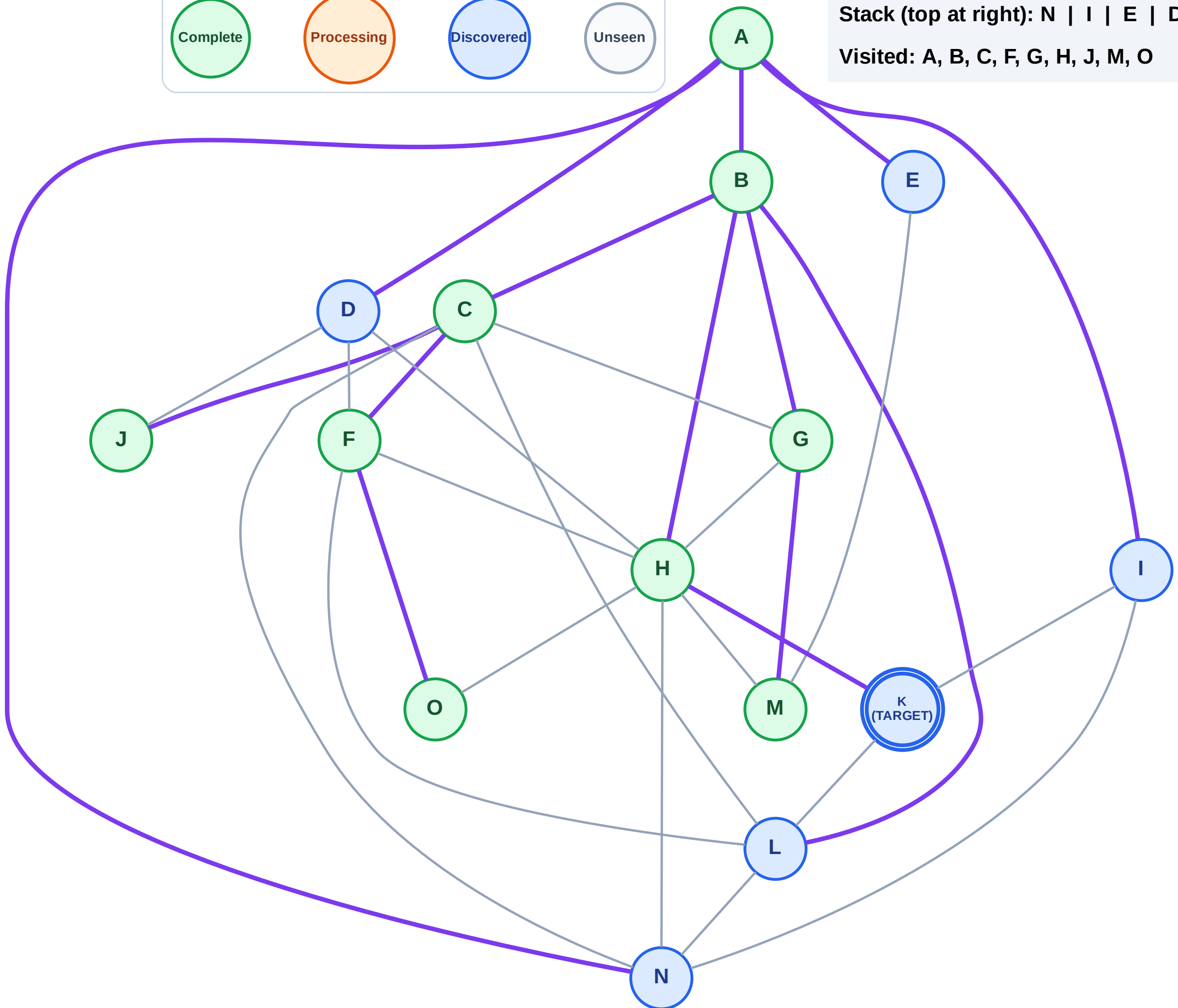
Step 19: Discover K from H

Legend



Stack (top at right): N | I | E | D | L | K

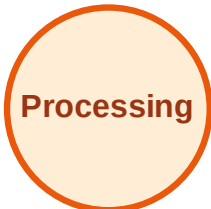
Visited: A, B, C, F, G, H, J, M, O



DFS Traversal

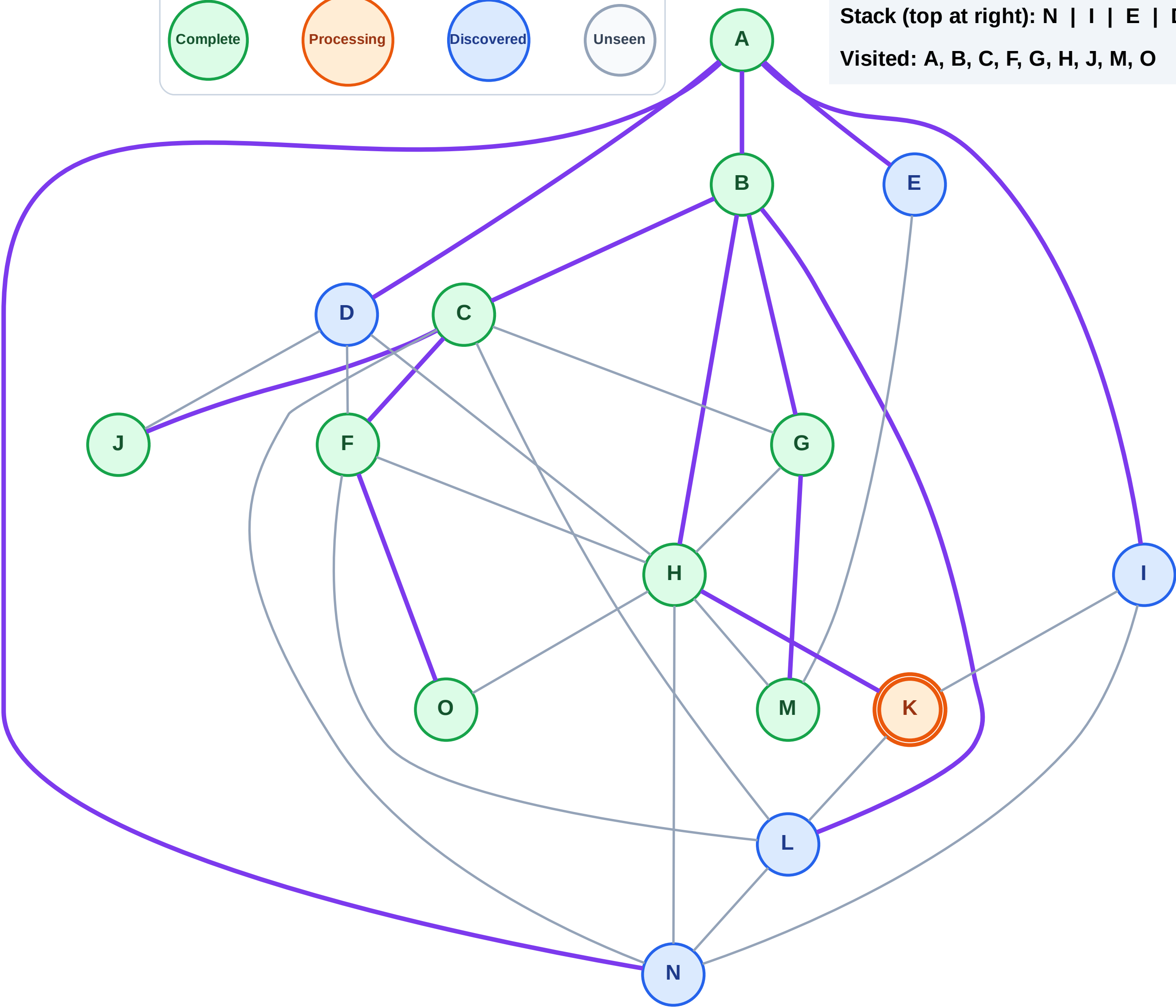
Step 20: Process node K

Legend



Stack (top at right): N | I | E | D | L

Visited: A, B, C, F, G, H, J, M, O



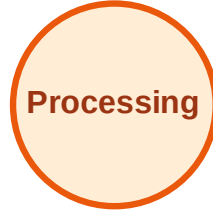
DFS Traversal

Step 21: TARGET NODE K REACHED! Path (3 hops): A → B → H → K

Legend



Complete



Processing



Discovered



Unseen

Stack (top at right): N | I | E | D | L

Visited: A, B, C, F, G, H, J, K, M, O

